Three-Layer Calibration for Bayesian Robust Variance Estimation in Meta-Analysis With Dependent Effect Sizes

JoonHo Lee and Elizabeth Tipton

Abstract. Meta-analyses in education, psychology, and medicine routinely extract several statistically dependent effect sizes from each study, and the within-study correlations needed to model that dependence are rarely reported. Frequentist practice resolves this through robust variance estimation: working models paired with cluster-robust standard errors, bias-reduced small-sample corrections, and Satterthwaite degrees of freedom. Bayesian meta-analysis offers complementary strengths (principled shrinkage, flexible priors on heterogeneity, and direct probability statements), but its posterior spread inherits whatever working dependence model the analyst assumes, and a Bayesian counterpart that calibrates this spread to the robust sandwich covariance has so far been lacking. This article introduces Three-Layer Calibration, a calibrated-Bayes procedure for Bayesian meta-regression with dependent effect sizes. Layer 1 retains the posterior mean as the point estimate. Layer 2 applies a Cholesky Calibration Correction that affinely transforms posterior draws so their sample covariance matches a bias-reduced (CR2) cluster sandwich exactly in finite samples while preserving the posterior mean. Layer 3 replaces the normal reference distribution with per-coefficient Satterthwaite t distributions. In a 324-condition simulation built on a real meta-analysis, calibrated intervals attained 0.952 coverage for the average effect and uniform 0.966–0.978 across slopes, which carry roughly one third the intercept's degrees of freedom. A normal-reference variant undercovered all three slopes (0.908–0.931); the frequentist benchmark undercovered only the smallest (0.905), where the largest gain occurs, at a documented power cost. A reanalysis of 1,198 effect sizes from 117 studies of brief alcohol interventions illustrates the workflow. Methods are implemented in the R package bayesRVE.

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1. Introduction

Most meta-analyses do not analyze one effect size per study. Primary studies contribute estimates for multiple outcome measures, multiple treatment contrasts, multiple subgroups, or multiple follow-up times, so that the resulting effect sizes are statistically dependent within studies, and the sampling correlations that generate this dependence are almost never reported (Hedges et al., 2010; López-López et al., 2018; Tanner-Smith et al., 2016). Ignoring the dependence, that is, treating N correlated effect sizes as independent, overstates the effective amount of information, understates standard errors, and yields confidence intervals and tests that are too liberal.

Frequentist research synthesis has converged on a mature solution to this problem. Robust variance estimation (RVE), introduced to meta-analysis by Hedges et al. (2010), pairs a deliberately simplified *working model* for the dependence structure with a cluster-robust (sandwich) variance estimator whose validity does not depend on the working model being correct. A sequence of refinements has made the approach practical at realistic scale: small-sample corrections that remove the downward bias of the sandwich estimator (Bell & McCaffrey, 2002; Pustejovsky & Tipton, 2018; Tipton, 2015; Tipton & Pustejovsky, 2015), Satterthwaite degrees of freedom that correct the calibration of tests when the number of studies is modest (Satterthwaite, 1946; Tipton, 2015), working models such as correlated effects (CE) and correlated-and-hierarchical effects (CHE) that represent plausible dependence structures parsimoniously (Pustejovsky & Chen, 2024; Pustejovsky & Tipton, 2022), and software that makes these methods routine to apply (Joshi et al., 2022; Tanner-Smith & Tipton, 2014; Vembye et al., 2024; Viechtbauer, 2010). The resulting standard of care, a CHE-type working model combined with a CR2-corrected sandwich and Satterthwaite degrees of freedom, is the established small-sample-corrected robust benchmark for coefficient-level inference. This benchmark is asymptotically robust in the number of studies, the CR2 and Satterthwaite corrections improve its finite-sample calibration, and it continues to treat the unknown within-study correlation as a nuisance rather than as an estimation target.

Bayesian meta-analysis has developed alongside this line of work, and it has several attractive features. Hierarchical Bayesian models estimate heterogeneity with principled uncertainty quantification even when the number of studies is small (Gelman, 2006; D. R. Williams et al., 2018), extend naturally to multilevel and multivariate dependence structures (Fernández-Castilla et al., 2021; Jackson et al., 2011; Van Den Noortgate et al., 2015), accommodate prior information and shrinkage where the data are weak, and yield direct probability statements about parameters, with increasing attention to their frequentist properties (Bartoš et al., 2026; Mathur, 2025). However, Bayesian methods handle dependent effect sizes by modeling them, and this creates a difficulty. The posterior standard deviation of a meta-regression coefficient is computed under the assumed working covariance. When the working model is misspecified, which is the usual situation because the true within-study correlations are unknown, the posterior spread describes the uncertainty that would be appropriate if the working model were true rather than the variability of the estimator under repeated sampling. A frequentist analyst who uses the same working model is protected by the sandwich estimator, whereas a Bayesian analyst has no analogous protection. As far as we are aware, no published Bayesian meta-analysis method takes the RVE sandwich covariance as an explicit calibration target for the posterior.

This gap can be closed with two ingredients that already exist. The first is the calibrated-Bayes perspective articulated by Rubin (1984) and Little (2012), which holds that Bayesian procedures should be evaluated, and where needed adjusted, on the basis of their frequentist operating characteristics. The second is a computational mechanism developed in survey statistics, an area in which a formally analogous problem arises when clustered observations are analyzed under convenient working likelihoods. Building on the asymptotic theory of the Bayesian pseudo-posterior under informative sampling (Savitsky & Toth, 2016), Williams and Savitsky developed a post-processing calibration in which the Monte Carlo draws are transformed so that their covariance agrees with a design-consistent sandwich estimate (Savitsky & Williams, 2022; M. R. Williams & Savitsky, 2021). That work is restricted to the survey setting and is asymptotic in character, and before the present two-article program it had not been connected to research synthesis. What has been lacking is the identification of an appropriate calibration target for meta-analysis, namely the CR2 cluster sandwich that the RVE literature has already made canonical, together with an account of the small-sample tail behavior that meta-analytic data require. A companion article (Lee et al., 2026) supplies the population-level foundation for this connection: under matched conditions, a dependent-effects meta-analysis and a clustered Bayesian survey analysis are the same model under two interpretations, so the covariance that a calibrated Bayesian meta-analysis should report is identified with the marginal fixed-effects sandwich that the survey framework already targets.

In this article we make this connection operational. We propose *Three-Layer Calibration* for Bayesian meta-regression with dependent effect sizes:

> **Layer 1 (location).** The posterior mean of the meta-regression coefficients is retained as the point estimate, so that the shrinkage and pooling benefits of the Bayesian fit are preserved.
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> **Layer 2 (spread).** A *Cholesky Calibration Correction* (CCC), an affine transformation of the posterior draws, is applied. The transformation leaves the posterior mean fixed and makes the sample covariance of the draws match a bias-reduced CR2 cluster-sandwich target exactly. The matching is an algebraic identity in finite samples rather than an approximation.
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> **Layer 3 (tails).** Interval estimates are constructed from the calibrated draws using per-coefficient Satterthwaite degrees of freedom, so that the normal reference distribution is replaced by t distributions that account for the variability of the sandwich estimator itself.

Each layer has a distinct epistemic status, and we keep the layers separate throughout. The covariance-matching property of Layer 2 is guaranteed. We state exact finite-sample propositions and verify them to machine precision. Whether the calibrated intervals attain nominal frequentist coverage is a different question, one that is asymptotic in the number of studies and dependent on the quality of the CR2 target. We answer this question empirically, using a 324-condition simulation built on a real meta-analytic database and a full empirical reanalysis. The simulation produces a result that we did not fully anticipate. The calibration matters most for the meta-regression slopes rather than for the average effect, for which several methods perform adequately. For the slopes, the proposed intervals maintain coverage of 0.966–0.978 across coefficients. The same calibrated draws evaluated against a normal reference undercover all three slopes (0.908–0.931), and the frequentist CR2 benchmark is coefficient-dependent, with coverage near or above the nominal level for two slopes but 0.905 for the smallest slope. The largest gain therefore occurs where effective degrees of freedom are fewest. The mechanism can be traced directly, since the slopes carry roughly one third the Satterthwaite degrees of freedom of the intercept. The same simulation also quantifies the costs of the approach. Power is sharply reduced for very small slopes, interval widths increase where identification is weakest, and the calibrated standard errors are mildly conservative. On the substantial slope, by contrast, the calibrated intervals are narrower than those of the CR2 benchmark and the point estimate has a smaller root mean squared error, reflecting the shrinkage that Layer 1 retains.

The plan of the article is as follows. In Section 2 we fix notation, review the two inferential traditions, and locate the gap. In Section 3 we develop the three layers, state the exact-matching propositions, and describe the implementation in the R package `bayesRVE`. In Section 4 we report the simulation. In Section 5 we reanalyze the Tanner-Smith and Lipsey (2015) meta-analysis of brief alcohol interventions (117 studies, 1,198 effect sizes). Finally, in Section 6 we discuss scope, limitations, and extensions. An online supplement, included with this article as Appendices A–G, provides complete derivations, proofs, simulation details, and additional results.

2. Dependent effect sizes and two inferential traditions

2.1. Data structure and working models. Study $j = 1, \ldots, J$ contributes n_j effect size estimates y_{ij}, $i = 1, \ldots, n_j$, with $N = \sum_j n_j$ in total. Each estimate carries a known sampling variance v_{ij}, and $\bar{s}_j^2 = n_j^{-1} \sum_i v_{ij}$ denotes the study average. A K-vector of moderators $\mathbf{x}_{ij}$ (including the intercept) is observed for each estimate; $\mathbf{X}_j$ is the $n_j \times K$ design matrix of study j, and $\bar{\mathbf{x}}_j$ is its column mean. The estimates within a study are dependent because they share participants, procedures, and constructs. The sampling correlation ρ that quantifies this dependence is a conditional correlation among the sampling errors, to be distinguished from the marginal correlation induced by shared random effects, and it is rarely reported in primary studies.

We consider two working models from the RVE tradition (Hedges et al., 2010; Pustejovsky & Tipton, 2022). The *correlated effects* (CE) working model specifies a study-level random effect,

$$y_{ij} = \mathbf{x}_{ij}^\top \boldsymbol{\beta} + \theta_j + \varepsilon_{ij}, \qquad \theta_j \sim \mathrm{N}(0, \tau^2), \qquad \varepsilon_{ij} \sim \mathrm{N}(0, \bar{s}_j^2), \tag{2.1}$$

where $\boldsymbol{\beta} \in \mathbb{R}^K$ collects the meta-regression coefficients, τ^2 is the between-study variance, and, in accordance with the CE convention, the within-study sampling variance is summarized by its study average $\bar{s}_j^2$. The *correlated-and-hierarchical effects* (CHE) working model adds an effect-level random component $w_{ij} \sim \mathrm{N}(0, \omega^2)$, so that heterogeneity is partitioned into between-study (τ^2) and within-study (ω^2) sources. When the estimates of study j are stacked, the implied working covariance of $\mathbf{y}_j$ is compound symmetric,

$$\boldsymbol{\Sigma}_j \;=\; a_j \mathbf{I}_{n_j} + \tau^2 \mathbf{1}_{n_j}\mathbf{1}_{n_j}^\top, \qquad a_j = \begin{cases} \bar{s}_j^2 & \text{(CE)} \\ \bar{s}_j^2 + \omega^2 & \text{(CHE)}, \end{cases} \tag{2.2}$$

and the study mean satisfies $\bar{y}_j \sim \mathrm{N}(\bar{\mathbf{x}}_j^\top \boldsymbol{\beta},\, v_j)$ with $v_j = a_j/n_j + \tau^2$.

Both specifications are working models in the sense that they are deliberate simplifications, adopted because the true within-study covariance is unknown rather than because they are believed to be correct. In particular, neither model attempts to estimate the sampling correlation among the ε_{ij}; frequentist implementations typically impute a constant working value (e.g., $\rho = 0.6$ or 0.8) when a full covariance matrix is required (Pustejovsky & Tipton, 2022), and assessment of sensitivity to that choice is part of standard practice (Tipton et al., 2019).

2.2. **The frequentist solution: robust variance estimation.** In the frequentist approach, the working model is required only to supply reasonable weights; the validity of the standard errors does not rest on its correctness. Writing $\mathbf{W}_j = \boldsymbol{\Sigma}_j^{-1}$ for the working weight matrices and $\hat{\boldsymbol{\beta}}$ for the weighted least squares (WLS) estimator, the cluster-robust sandwich estimator

$$\mathbf{V}_{\mathrm{CR0}} \;=\; \mathbf{Q}\Big(\textstyle\sum_{j=1}^J \mathbf{s}_j \mathbf{s}_j^\top\Big)\mathbf{Q}, \qquad \mathbf{s}_j = \mathbf{X}_j^\top \mathbf{W}_j \hat{\mathbf{e}}_j, \qquad \mathbf{Q} = \Big(\textstyle\sum_{j=1}^J \mathbf{X}_j^\top \mathbf{W}_j \mathbf{X}_j\Big)^{-1}, \tag{2.3}$$

with $\hat{\mathbf{e}}_j = \mathbf{y}_j - \mathbf{X}_j \hat{\boldsymbol{\beta}}$, is consistent for the true sampling variance of $\hat{\boldsymbol{\beta}}$ as $J \to \infty$ under essentially arbitrary misspecification of the within-study dependence (Cameron & Miller, 2015; Hedges et al., 2010). Because the middle factor (the "meat") in (2.3) accumulates one score per independent cluster, the robustness derives from the number of studies rather than from the number of effect sizes.

Two small-sample corrections complete the approach. First, the residuals $\hat{\mathbf{e}}_j$ are systematically too small, because the fitted model absorbs part of each study's error, so (2.3) is biased downward when J is modest. The CR2 correction (Bell & McCaffrey, 2002; Imbens & Kolesár, 2016), extended to weighted meta-regression by Pustejovsky and Tipton (2018), Tipton (2015), and Tipton and Pustejovsky (2015), inflates each residual by a study-specific adjustment matrix $\mathbf{A}_j$ chosen to remove the leading small-sample bias of the sandwich under the working model (Section 3.2). Second, even an unbiased variance estimate is itself noisy, and comparing t statistics to a normal reference produces liberal tests. Satterthwaite degrees of freedom (Satterthwaite, 1946; Tipton, 2015), computed per coefficient from the variance of the variance estimator, restore test calibration; simulation evidence consistently shows that with fewer than roughly 40 studies, and especially with unbalanced or skewed moderators, the CR2+Satterthwaite combination is required to keep rejection rates near their nominal levels (Joshi et al., 2022; Tipton, 2015; Tipton & Pustejovsky, 2015; Welz et al., 2023). These corrections should not be regarded as optional refinements, since uncorrected RVE is known to be anticonservative in small meta-analyses; statements that RVE performs well at small J therefore implicitly refer to a corrected version such as CR2 (Tipton, 2015; Tipton & Pustejovsky, 2015).

We refer to the complete procedure, consisting of a CHE-type working model, the CR2-corrected sandwich, and a per-coefficient Satterthwaite t reference, as the *frequentist standard of care* for dependent effect sizes. It is instantiated in the R packages `robumeta`, `metafor`, and `clubSandwich` (Fisher et al., 2023; Pustejovsky, 2026; Viechtbauer, 2010), and it is the benchmark that any Bayesian analogue must meet.

2.3. **Bayesian meta-analysis and the inferential mismatch.** A Bayesian analyst faced with the same data places priors on $(\boldsymbol{\beta}, \tau, \omega)$, fits the CE or CHE working model by Markov chain Monte Carlo (MCMC), and reports posterior summaries. The posterior mean of $\boldsymbol{\beta}$ performs well as a point estimate. It incorporates partial pooling, stabilizes small-study estimates, and, under mild regularity conditions, remains consistent even when the working covariance is misspecified, since the working-model maximum likelihood estimand coincides with the WLS estimand to which the sandwich applies. The difficulty concerns the posterior spread. Under misspecification, the asymptotic variance of the posterior mean is the sandwich, whereas the posterior variance converges to the inverse Fisher information of the working model, and in general the two do not agree (Müller, 2013). Writing $\mathrm{SD}_{\mathrm{post}}(\beta_k)$ for the posterior standard deviation and $\mathrm{SE}_{\mathrm{CR2}}(\beta_k)$ for the robust standard error, the mismatch ratio

$$M_k \;=\; \frac{\mathrm{SD}_{\mathrm{post}}(\beta_k)}{\mathrm{SE}_{\mathrm{CR2}}(\beta_k)} \tag{2.4}$$

equals one only when the working model is correctly specified. Three features of meta-analytic data make $M_k \neq 1$ practically consequential: working-model misspecification is the norm (the sampling correlations are unknown); J is often small, so that the prior and the working likelihood dominate the spread; and cluster sizes n_j are typically very unbalanced, which concentrates leverage in a manner that the working model does not represent.

It is worth being precise about how this failure manifests itself, because it is easy to overstate. In our simulation (Section 4), uncalibrated posterior intervals from a correctly specified random-effects structure attain coverage close to the nominal level on average. The deterioration arises where it cannot be diagnosed from within the model. Under a misspecified working structure, condition-level coverage of the uncalibrated interval for the average effect falls as low as 0.818, and it worsens as J grows (0.843 at $J = 20$, 0.826 at $J = 40$, and 0.818 at $J = 80$ in the worst cells), because with more data the posterior spread is determined increasingly by the misspecified working model. The calibration of a Bayesian interval therefore depends on an assumption that the analyst cannot check, whereas the calibration of the frequentist interval, which is based on the sandwich, does not. Bayesian meta-analysis has so far lacked an analogous form of protection.

One response to this problem is to model the dependence more fully rather than to protect against it, for instance by placing priors on the unreported within-study correlations or by fitting a multivariate dependence model that treats the correlation structure as an estimation target. That approach is valuable when the data can identify those correlations or when credible external information about them exists. This article addresses the RVE setting, in which neither condition can be relied upon. In that setting, the dependence structure is a nuisance, often weakly identified from the extracted effect sizes and sometimes not identified at all, and the target is coefficient-level inference that is robust to this nuisance. Calibration is therefore not a substitute for a correct full likelihood; it is a reporting layer for the common case in which that likelihood cannot be known.

2.4. **Calibrated Bayes and the survey-calibration mechanism.** The approach we develop is an application of an established inferential position rather than a new one. Rubin (1984) argued that Bayesian procedures should be evaluated by their frequentist operating characteristics, and Little (2012) termed the resulting position *calibrated Bayes*, under which inference proceeds from the posterior but the reported uncertainty should be trustworthy in repeated sampling. From this standpoint, adjusting the posterior spread to match a defensible frequentist target does not conflict with Bayesian principles; it serves as a form of quality control on the working model.

The computational mechanism comes from survey statistics, where a formally parallel problem (independent clusters, convenient working likelihoods, misspecified dependence) arises under informative sampling. Incorporating the survey design into a working likelihood has a long history in that field, from pseudo-likelihood estimation with sandwich standard errors (Binder, 1983; Pfeffermann et al., 1998) to pseudo-maximum-likelihood methods for multilevel models (Rabe-Hesketh & Skrondal, 2006). Savitsky and Toth (2016) established asymptotic theory for the corresponding Bayesian pseudo-posterior under informative sampling. Williams and Savitsky then showed that pseudo-posteriors understate design variance and proposed a post-processing adjustment in which the MCMC draws are recentered and rescaled, using Cholesky factors of the two covariance matrices, so that their covariance matches a design-consistent sandwich estimate (Savitsky & Williams, 2022; M. R. Williams & Savitsky, 2021). Their results are asymptotic and confined to the survey setting, and their transformation had not been applied in research synthesis before the present program; the companion article (Lee et al., 2026) proves the likelihood identity that carries the framework across and establishes the exact center-preserving form of the map at the population level. Several general-purpose proposals share this diagnosis. The "open-faced sandwich" adjustment of Shaby (2014) transforms the draws to target the Godambe (composite-likelihood) covariance of the working estimator rather than the meta-analytic CR2 target, while generalized and loss-based posteriors (Bissiri et al., 2016; Frazier et al., 2026; Lyddon et al., 2019) and composite-likelihood corrections (Varin et al., 2011) leave the meta-analytic question open, because none of them identifies which covariance a calibrated meta-analytic posterior should match. The most closely related work within research synthesis is the calibrated Bayesian composite-likelihood method of Wang et al. (2025) for multivariate network meta-analysis, which applies the same open-faced-sandwich post-sampling adjustment when within-study correlations are unreported. Its target is again the Godambe covariance of that composite-likelihood estimator; it neither identifies the CR2 covariance of the RVE tradition as the calibration target for meta-regression with dependent effect sizes nor supplies per-coefficient Satterthwaite small-sample tails. The estimating-equations tradition (Binder, 1983), read together with the RVE methods of Section 2.2, supplies the appropriate target, namely the bias-reduced CR2 cluster sandwich evaluated under the same working model that the Bayesian fit uses. The contribution of the present article is the finite-sample operational method: the bias-reduced CR2 instantiation of this target, the survey-style transformation of the draws that attains it exactly, the per-coefficient Satterthwaite tail layer that meta-analysis requires, and the simulation and empirical evaluation of the result. We call the resulting transformation of the draws the *Cholesky Calibration Correction*

(CCC); this is our label for the meta-analytic instantiation of the Williams–Savitsky scale-and-shape mechanism.

3. Three-Layer Calibration

The procedure operates entirely after model fitting: the analyst fits a standard Bayesian CE or CHE model once and then applies the calibration. Throughout, $\boldsymbol{\psi}^{(s)}$, $s = 1, \ldots, S$, denotes the MCMC draws of the parameter vector, and $\hat{\boldsymbol{\psi}} = S^{-1} \sum_s \boldsymbol{\psi}^{(s)}$ denotes the posterior mean. For the CE model $\boldsymbol{\psi} = (\boldsymbol{\beta}^\top, \theta_1, \ldots, \theta_J)^\top$; for the CHE model, which integrates the study effects out of the likelihood, the calibrated block is $\boldsymbol{\beta}$ alone.

3.1. **Layer 1: Location from the posterior mean.** The point estimate is the posterior mean $\hat{\boldsymbol{\beta}}$. Calibration does not require replacing this estimate: the affine map of Layer 2 is constructed to hold the posterior mean fixed (Proposition 3.1), so the shrinkage, pooling, and prior regularization that motivated the Bayesian fit remain intact after calibration. Note that the fixed point of the calibration is the posterior mean, that is, the Monte Carlo average of the draws, and not the posterior mode.

3.2. **The calibration target: a CR2 cluster sandwich.** Layer 2 requires a target covariance for $\hat{\boldsymbol{\beta}}$, and the frequentist RVE literature indicates what this target should be. Working at plug-in values of the variance components (posterior means $\hat{\tau}^2$ and, for CHE, the Jensen-consistent $\hat{\omega}^2 = S^{-1} \sum_s (\omega^{(s)})^2$), we form the working covariance blocks $\boldsymbol{\Sigma}_j$ of (2.2), the weights $\mathbf{W}_j = \boldsymbol{\Sigma}_j^{-1}$, the WLS bread $\mathbf{Q}$ as in (2.3), and the residuals $\hat{\mathbf{e}}_j = \mathbf{y}_j - \mathbf{X}_j \hat{\boldsymbol{\beta}}$ at the posterior mean. The CR2 adjustment matrices are

$$\mathbf{A}_j \;=\; \mathbf{W}_j^{-1/2} \big(\mathbf{I}_{n_j} - \tilde{\mathbf{H}}_{jj}\big)^{-1/2} \mathbf{W}_j^{1/2}, \qquad \tilde{\mathbf{H}}_{jj} = \mathbf{W}_j^{1/2} \mathbf{X}_j \mathbf{Q} \mathbf{X}_j^\top \mathbf{W}_j^{1/2}, \tag{3.1}$$

where the symmetrized hat matrix $\tilde{\mathbf{H}}_{jj}$ measures the leverage of study j. The adjusted scores and the target are

$$\tilde{\mathbf{s}}_j = \mathbf{X}_j^\top \mathbf{W}_j \mathbf{A}_j \hat{\mathbf{e}}_j, \qquad \mathbf{V}_{\mathrm{CR2}} \;=\; \mathbf{Q} \Big(\textstyle\sum_{j=1}^J \tilde{\mathbf{s}}_j \tilde{\mathbf{s}}_j^\top \Big) \mathbf{Q}. \tag{3.2}$$

The adjustment $\mathbf{A}_j$ inflates the residual of each study so that $\mathbb{E}[\mathbf{V}_{\mathrm{CR2}}]$ equals the true variance of $\hat{\boldsymbol{\beta}}$ under the working model, up to a cross-study term of order $1/J$ (Appendix B); this removes the leading downward bias of (2.3) in small samples (Bell & McCaffrey, 2002; Tipton, 2015); its robustness to misspecification of the dependence structure is inherited from the sandwich form. Three implementation choices deserve comment. First, the bread is the WLS matrix $\mathbf{Q}$ rather than the posterior

Hessian: CR2 is a hat-matrix construction, and introducing prior curvature into $\mathbf{Q}$ would distort the leverage computations on which the exactness of the correction depends. Second, the compound-symmetric structure of $\boldsymbol{\Sigma}_j$ admits closed-form $\mathbf{W}_j^{1/2}$ and $\mathbf{W}_j^{-1/2}$, so (3.1) costs $O(n_j)$ rather than $O(n_j^3)$ (Appendix B). Third, the target is conditional on the plug-in variance components: if $\hat{\omega}^2$ is severely misestimated, the target itself shifts, a caveat to which we return in Section 6. Our implementation reproduces `clubSandwich`'s CR2 and Satterthwaite computations to within 10^{-6} when supplied with identical plug-ins (Appendix D).

In a Bayesian workflow, $\mathbf{V}_{\mathrm{CR2}}$ should be read as a calibrated-Bayes reporting target rather than as an approximation to the posterior Hessian. The prior enters through the reported location $\hat{\boldsymbol{\beta}}$ and through the variance-component plug-ins, whereas the spread target is score-based by design: it is the CR2 covariance of the working-model estimating equations at those plug-ins, built on the WLS bread $\mathbf{Q}$. Layer 2 therefore addresses a deliberately frequentist question, namely how variable the reported coefficient estimator is under clustered repeated sampling, while Layer 1 retains the Bayesian location. The result is a calibrated reporting distribution for $\hat{\boldsymbol{\beta}}$; we do not claim that the posterior covariance equals the sampling variance of a fully Bayesian estimator under all priors. The conditionality noted above is intrinsic to this reading: if the plug-ins are unstable, the target itself moves, and the calibration diagnostics of Section 3.3 and the sensitivity analyses of Section 4 should be interpreted accordingly.

For the CE model, which carries explicit study effects θ_j, the same logic yields per-study conditional sandwich variances for the θ_j (scalar scores and Hessians given $\hat{\boldsymbol{\beta}}$), which are collected in a diagonal matrix $\mathbf{V}^{(\theta)}$. No cross-block target is defined, for a structural reason that we describe below.

3.3. **Layer 2: Cholesky calibration to the target.** Let $\hat{\boldsymbol{\Sigma}}_{\mathrm{MCMC}}$ denote the sample covariance of the draws of a parameter block, and $\mathbf{V}_{\mathrm{target}}$ the corresponding target block. We take lower-triangular Cholesky factors $\mathbf{L}_1 = \mathrm{chol}(\hat{\boldsymbol{\Sigma}}_{\mathrm{MCMC}})$ and $\mathbf{L}_2 = \mathrm{chol}(\mathbf{V}_{\mathrm{target}})$ and transform every draw by the affine map

$$\boldsymbol{\psi}^{*(s)} \;=\; \hat{\boldsymbol{\psi}} + \mathbf{A}_{\mathrm{ccc}}\big(\boldsymbol{\psi}^{(s)} - \hat{\boldsymbol{\psi}}\big), \qquad \mathbf{A}_{\mathrm{ccc}} = \mathbf{L}_2\mathbf{L}_1^{-1}, \qquad s = 1, \ldots, S. \tag{3.3}$$

Geometrically, $\mathbf{L}_1^{-1}$ standardizes the cloud of posterior draws to a sphere and $\mathbf{L}_2$ restandardizes it to the target ellipsoid; the map rotates and rescales the spread and has no other effect. Three algebraic facts govern the transformation; the proofs are given in Appendix C.

Proposition 3.1 (Center preservation)**.** *The calibrated draws have sample mean* $\bar{\boldsymbol{\psi}}^* = \hat{\boldsymbol{\psi}}$ *exactly, for any* $\mathbf{A}_{\text{ccc}}$ *of the form* (3.3)*.*

Proposition 3.2 (Exact covariance matching)**.** *The calibrated draws have sample covariance* $\text{Cov}(\boldsymbol{\psi}^{*(1)}, \ldots, \boldsymbol{\psi}^{*(S)}) = \mathbf{V}_{\text{target}}$ *exactly—an algebraic identity holding for every S exceeding the block dimension, every draw configuration with positive-definite* $\hat{\boldsymbol{\Sigma}}_{\text{MCMC}}$*, and every positive-definite target, with no distributional or asymptotic input.*

Proposition 3.3 (No-op under exact covariance matching)**.** *If* $\hat{\boldsymbol{\Sigma}}_{\text{MCMC}} = \mathbf{V}_{\text{target}}$*, then* $\mathbf{A}_{\text{ccc}} = \mathbf{I}$ *and every draw is returned unchanged.*

Proposition 3.2 is the central guarantee: the covariance-matching property is an algebraic identity rather than an approximation, and it is verified in the implementation to machine precision (deviations of order 10^{-13}). Proposition 3.3 is a benchmark rather than a certificate of correctness: it states that if the draw covariance already equals the target, CCC returns the draws unchanged. Under a well-specified working model with weak priors the calibration map is expected to approach the identity as J grows, but correct specification does not make the finite-sample draw covariance equal the CR2 target, because the small-sample CR2 adjustment and the plug-in variance components can still induce a nontrivial calibration. The case for applying CCC by default rests on the benchmark alone: if the draw covariance already equals the target, the transformation leaves the draws unchanged, so the default application carries no cost in that case. Proposition 3.1 ensures that the Layer 1 location is preserved.

One structural decision remains, namely which parameter blocks to calibrate. The full CE parameter vector has dimension $K + J$, but the full-joint sandwich meat is a sum of J rank-one score outer products, so its rank is at most $J < K + J$. The full-joint target is thus structurally singular, an expression of the incidental-parameters geometry (Neyman & Scott, 1948), and admits no Cholesky factor. CCC is therefore blockwise by design: the $\boldsymbol{\beta}$ block is calibrated with the full $K \times K$ matrix map (3.3) against $\mathbf{V}_{\text{CR2}}$, and each θ_j is rescaled against its scalar conditional target, $\theta_j^{*(s)} = \hat{\theta}_j + r_j(\theta_j^{(s)} - \hat{\theta}_j)$ with $r_j = (V_{jj}^{(\theta)} / \hat{\Sigma}_{\text{MCMC},jj}^{(\theta)})^{1/2}$. The block split is not a computational workaround; the two blocks answer different inferential questions (population-level coefficients versus study-specific effects), and the marginal fixed-effects target is selected by the reporting estimand (the quantity that a meta-analysis reports averages over the study effects), independently of the rank obstruction; the companion article gives the formal treatment of both the obstruction and the estimand-based target selection (Lee et al., 2026). For the CHE model, whose likelihood already integrates the θ_j out, only the $\boldsymbol{\beta}$ block exists and only this block is calibrated.

The per-coefficient ratios $r_k = \{[\mathbf{V}_{\mathrm{CR2}}]_{kk}/[\hat{\boldsymbol{\Sigma}}_{\mathrm{MCMC}}]_{kk}\}^{1/2}$ and the deviation $\|\mathbf{A}_{\mathrm{ccc}} - \mathbf{I}_K\|_F$ serve as diagnostics that require no additional computation: they quantify, coefficient by coefficient, the distance between the working-model posterior spread and the robust target.

3.4. **Layer 3: Satterthwaite reference distributions.** Matching the covariance is necessary but not sufficient for calibrated intervals, because $\mathbf{V}_{\mathrm{CR2}}$ is itself an estimate with nontrivial sampling variability when J is small. Frequentist RVE addressed this problem with per-coefficient Satterthwaite degrees of freedom (Pustejovsky & Tipton, 2018; Satterthwaite, 1946; Tipton, 2015), and Layer 3 adopts that solution without modification. For coefficient k, the degrees of freedom take the standard form

$$\mathrm{df}_k \;=\; \frac{\big[\mathrm{tr}(\mathbf{P}_k)\big]^2}{\mathrm{tr}(\mathbf{P}_k^2)}, \tag{3.4}$$

where $\mathbf{P}_k$ is the $J \times J$ matrix assembled from the CR2-adjusted projection vectors $\mathbf{g}_{jk} \;=\; \mathbf{A}_j^\top \mathbf{W}_j \mathbf{X}_j \mathbf{Q}\,\mathbf{c}_k$ and the working covariance blocks, evaluated at the plug-in variance components ($\mathbf{c}_k$ is the kth standard basis vector; the construction is given in Appendix D). The $100(1-\alpha)\%$ calibrated interval, which we call the *CCC-t* interval, is then

$$\hat{\beta}_k \;\pm\; t_{1-\alpha/2,\,\mathrm{df}_k}\;\mathrm{SD}\big(\beta_k^*\big), \tag{3.5}$$

where $\mathrm{SD}(\beta_k^*)$ is the standard deviation of the calibrated draws. The df in (3.4) are computed separately for each coefficient, and this feature is more consequential than is generally recognized: in realistic meta-regression designs the intercept can carry two to three times the degrees of freedom of the moderator slopes, because the information about the slopes is concentrated in the between-study contrasts of a few clusters. Section 4.3 shows that this asymmetry in the df is what distinguishes methods that attain nominal coverage for the slopes from methods that do not. We use CCC-q ("quantile") to denote the variant that retains Layers 1–2 but reads the interval endpoints from the calibrated draws with a normal reference (df $= \infty$); because the two methods share identical calibrated draws, a comparison of CCC-t with CCC-q isolates the contribution of the tail layer.

3.5. **What is guaranteed and what is demonstrated.** The claims of this article fall into three strata, which we distinguish explicitly. The first stratum is guaranteed. The calibrated draws match the CR2 target covariance exactly and preserve the posterior mean exactly (Propositions 3.1–3.2); these are finite-sample algebraic identities, verified to machine precision, and they hold whenever the calibrated block has positive-definite source and target covariances and a sufficient number of posterior draws ($S > K$). The implementation screens for these conditions through the operating envelope $J \geq \max(8,\, 2K + 2)$ (Section 3.6) and through positive-definiteness checks, and it skips and flags non-positive-definite targets rather than repairing them silently (Appendix C). The second stratum is demonstrated rather than guaranteed. Intervals built from the calibrated draws attained near-nominal frequentist coverage under the data-generating processes we study (Sections 4–5); coverage is an operating characteristic that is asymptotic in J and depends in addition on the CR2 target being a good estimate of the truth. No algebraic identity can deliver such a property, and we do not claim it beyond the designs evaluated. The third stratum consists of claims that we deliberately do not make: we do not claim improved coverage over the frequentist standard of care in general, validity across dependence structures that we did not simulate, or robustness to a poorly estimated plug-in $\hat{\omega}^2$, which would shift the target itself.

3.6. **Implementation.** The procedure is implemented in the open-source R package `bayesRVE` (Lee & Tipton, 2026), with Stan as the sampling back end (Carpenter et al., 2017). A complete analysis consists of four function calls: `bm_fit()` fits the CE or CHE working model by MCMC (non-centered parameterization by default, half-t priors on τ and ω, weakly informative normal priors on $\boldsymbol{\beta}$); `sandwich()` constructs the CR2 target and Satterthwaite df at the posterior-mean plug-ins; `ccc()` applies the blockwise transformation (3.3); and `summary()` reports posterior means, calibrated standard deviations, per-coefficient df, CCC-t intervals, and the calibration diagnostics r_k and $\|\mathbf{A}_{\text{ccc}} - \mathbf{I}\|_F$. The cost of the calibration itself is negligible ($O(SK^2 + K^3)$, less than a millisecond at typical values $S = 8{,}000$ and $K \leq 12$), so the total cost is dominated by the MCMC fit. The package enforces an operating envelope $J \geq \max(8,\, 2K + 2)$ and refuses configurations in which no cluster-robust method (Bayesian or frequentist) has enough independent clusters for the sandwich target to be trustworthy; the refusal marks an applicability boundary rather than a convergence failure.

4. Simulation study

4.1. **Design.** We evaluated the operating characteristics in a factorial simulation of 324 conditions, constructed to be as faithful as practicable to real meta-analytic data.

Table 1. Simulation design. The crossed factors yield 324 retained conditions (282 core and balanced conditions with 1,000 replications; 42 sensitivity-block conditions with 500 replications).

Factor	Levels	Notes
Generating structure	CE; CHE	CHE adds within-study heterogeneity ω
Studies J	5, 10, 15, 20, 40, 80	envelope $J_{\min} = \max(8, 2K+2)$
Coefficients K	1, 4, 7	$K = 7$ requires $J \geq 16$ by design
Between-study SD τ	0.1, 0.2	brackets the empirical value 0.105
Within-study SD ω	0, 0.1, 0.2, 0.3	0 under CE only; 0.3 in sensitivity block
Sampling correlation ρ_{true}	0, 0.3, 0.8	0.6 in a sensitivity block
Cluster sizes n_j	empirical; balanced(5); balanced(1)	empirical profile capped at 12
Working correlation ρ_{work}	0.8	0.6 in sensitivity block (frequentist comparators)
Random-effect distribution	normal; contaminated	10% contamination, variance $\times 9$ (sensitivity)
Priors	default; narrow; diffuse	half-t_3 scale 0.5 / 0.1 / 2.0 (sensitivity)

Note. Moderators are resampled from the Tanner-Smith and Lipsey database; true coefficients are fixed at the values estimated from that database. Each generated dataset is analyzed by every estimator, so all comparisons are within-dataset.

Rather than specifying artificial covariate distributions, the data-generating process resamples study structures (cluster sizes, moderator values, and sampling-variance profiles) from the Tanner-Smith and Lipsey (2015) database of brief alcohol interventions as curated by Pustejovsky and Tipton (2022) (Section 5), generates standardized mean differences with genuinely correlated sampling errors (Wishart-based, with study-specific correlations drawn around ρ_{true}), and sets the true coefficients to the values estimated from that database. For the primary $K = 4$ design these values are $\beta_0 = 0.148$ (average effect), $\beta_1 = 0.031$ (a small slope), $\beta_2 = 0.290$ (a substantial slope), and $\beta_3 = 0$ (a null slope for Type-I error assessment). Table 1 summarizes the factors; Appendix E gives the full algorithm, the condition grid, and all design decisions.

Nine estimators were run on every dataset, forming five families: (i) *CCC-t*, the proposed three-layer procedure, in CE and CHE working-model variants; (ii) *CCC-q*, identical to CCC-t except that interval endpoints are read against a normal reference, so that the comparison isolates Layer 3; (iii) *uncalibrated Bayes*, the posterior 2.5%/97.5% credible interval from the same fits; (iv) *CR2*, the standard frequentist approach (`metafor` CE or CHE fit with sampling covariances imputed at ρ_{work}, `clubSandwich` CR2 with Satterthwaite df); and (v) *robumeta*, the original correlated-effects RVE with its small-sample correction, applicable to the intercept-only design. Crossing CE- and CHE-family estimators with CE and CHE generating structures yields correctly specified, overfitted, and misspecified evaluations of every method. Performance metrics follow standard simulation-reporting practice and comprise coverage of nominal 95% intervals (the primary metric), bias, interval width, root mean squared error, rejection rates, convergence, and Satterthwaite df, each accompanied by Monte Carlo standard errors (MCSEs). The main results pool conditions within the operating envelope; conditions in which a method does not run (the CE-arm refusal at $J = 5$, $K \geq 4$) are excluded for that method, and a worst-case sensitivity analysis that counts every such condition as a failure to cover is reported in Appendix E. Slope results are based on the 177 correctly specified $K \geq 4$ conditions (59 CE, 118 CHE), of which 176 lie inside the operating envelope and form the analyzed set.

4.2. **Interval coverage.** Table 2 reports the pooled coverage results; Figure 1 displays the slope results. For the average effect β_0, all four families perform adequately: CCC-t attains 0.952, CR2 0.947, uncalibrated Bayes 0.952, and CCC-q 0.936. However, the average effect is the least demanding estimand. It pools information from every study, carries the largest effective degrees of freedom, and is the estimand on which existing methods have been developed and evaluated most extensively.

The results for the moderator slopes differ markedly. On the small slope β_1, CR2, the standard frequentist approach with its own Satterthwaite correction, covers at 0.905, and CCC-q covers at 0.908, whereas CCC-t covers at 0.978. The costs of the CCC-t interval are documented in Section 4.4: its median width is three orders of magnitude larger than that of CR2, and its power is 0.027. On β_2 and β_3, the coverage of CR2 is satisfactory (0.961 and 0.955, at or above the nominal level), CCC-q remains below the nominal level (0.931 and 0.930), and CCC-t remains conservative (0.971 and 0.966); on β_2 its intervals are nevertheless narrower than those of CR2 (Section 4.4). The central finding is uniformity across coefficients rather than uniform

TABLE 2. Empirical coverage of nominal 95% intervals, pooled over correctly specified in-envelope conditions (unweighted mean of condition-level coverage). Pooled binomial Monte Carlo standard errors are at most 0.001 for every entry.

Estimand (true value)	CCC-t	CR2	CCC-q	Uncal. Bayes
β_0 average effect (0.148)	0.952	0.947	0.936	0.952
β_1 small slope (0.031)	**0.978**	0.905	0.908	0.961
β_2 larger slope (0.290)	**0.971**	0.961	0.931	0.960
β_3 null slope (0)	**0.966**	0.955	0.930	0.965

Note. CCC-t = Three-Layer Calibration (proposed); CR2 = frequentist working model with CR2-corrected sandwich and Satterthwaite t; CCC-q = calibrated draws with normal reference (isolates Layer 3); Uncal. Bayes = posterior credible interval from the same Bayesian fits. Bases: 319 in-envelope conditions for β_0; 176 in-envelope (of 177 designed) slope conditions for β_1–β_3. Entries are exact CE+CHE weighted pooled means; per-arm values are in Appendix E.

superiority: CCC-t maintains mildly conservative coverage of 0.966–0.978 across all three slopes, with the associated costs quantified below; the coverage of CR2 depends on the coefficient, ranging over 0.905–0.961, with clear undercoverage only on β_1; and CCC-q undercovers all three slopes (0.908–0.931).

The comparison between CCC-t and CCC-q isolates Layer 3. The two methods share the same calibrated spread and differ only in the reference distribution, and the intervals based on the normal reference undercover. It follows that calibrating the spread is not sufficient; the reference distribution must be calibrated as well. The numerical proximity of CCC-q (0.908) to CR2 (0.905) on β_1 provides empirical corroboration, from outside the Bayesian pipeline, that covariance calibration alone does not repair small-df tail behavior. It does not reflect an algebraic equivalence: CR2 differs in point estimate, target construction, and plug-ins, and already refers its statistic to Satterthwaite tails. The fact that its own t-correction nevertheless yields coverage of 0.905 indicates that estimation noise in the sandwich at df $\approx$ 4 undermines the t-correction itself.

Finally, uncalibrated Bayes appears satisfactory in Table 2 (0.960–0.965 on the slopes). However, as shown in Section 2.3, this average conceals the absence of protection: under working-model misspecification its condition-level coverage of β_0 falls to 0.818 in the worst cell and deteriorates as J grows, whereas CCC-t does not fall below 0.932 in the same cells. The uncalibrated interval is adequate when the working model is adequate, and the analyst cannot verify this condition in practice.

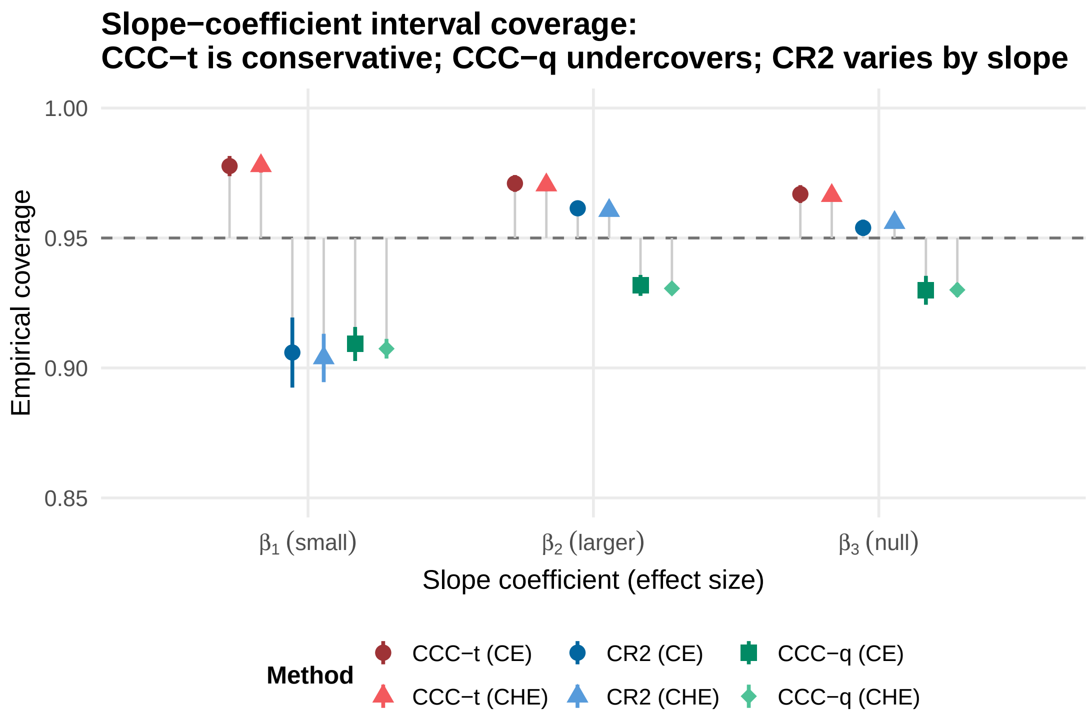

FIGURE 1. Coverage of nominal 95% intervals for the moderator slopes, by method and working-model arm (CE and CHE), pooled over the correctly specified slope conditions (176 in-envelope of the 177 designed). Points are unweighted means over conditions; bars span ± 1.96 between-condition standard errors; the dashed line marks 0.95. CCC-t remains at approximately 0.97 across all three slopes; CCC-q undercovers all three; CR2 undercovers most severely on the small slope β_1, which is its only clear undercoverage.

Coverage of CCC-t is also stable across the specification roles, by construction of the target: it is 0.978, 0.977, and 0.978 on β_1 under correct specification, overfitting, and misspecification, respectively, compared with 0.905–0.919 for CR2 (Figure 2). For the higher-order $K = 7$ moderators (β_4–β_6, 54 conditions), CCC-t and CR2 are statistically indistinguishable (0.952–0.961), and CCC-q is again lower (0.902–0.925); the advantage of calibration is concentrated where degrees of freedom are scarce rather than being uniform across settings.

4.3. **The Layer-3 mechanism: slopes carry one third the degrees of freedom.**

We now consider why the tail layer matters substantially more for the slopes than for the average effect. The explanation is visible in the Satterthwaite degrees of freedom themselves (Figure 3). Across the same 177 designed slope conditions, the median

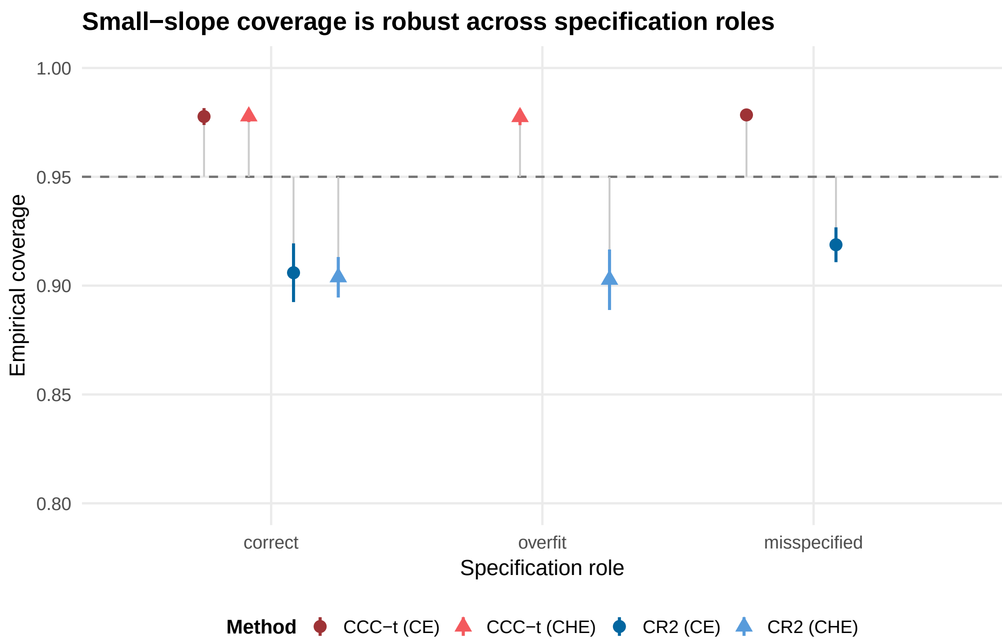

FIGURE 2. Coverage of the small slope β_1 by specification role. CCC-t (top) remains at 0.977–0.978 whether the working model is correctly specified, overfitted, or misspecified, reflecting the robustness inherited from the sandwich, whereas CR2 (bottom) ranges over 0.905–0.919. Bars span ± 1.96 between-condition standard errors; the dashed line marks 0.95.

per-condition df is 13.4 for β_0 but 4.3 for β_1, 4.9 for β_2, and 6.0 for β_3. The slopes therefore carry roughly one third of the effective degrees of freedom of the intercept, because information about a slope derives from between-study moderator contrasts that only a small number of clusters supply. At df $= 4.3$ the correct t multiplier for a 95% interval is 2.7 rather than 1.96, so that a normal-reference interval is approximately 28% too short regardless of how well its standard error is calibrated.

This mechanism yields a quantitative prediction: the CCC-t–CCC-q coverage gap should be ordered by the df deficit. This prediction holds across all four coefficients. The per-condition gap is 0.070 on β_1 (median df 4.3), 0.040 on β_2 (df 4.9), 0.037 on β_3 (df 6.0), and 0.016 on β_0 (df 13.4), so the gaps are monotone in the df ordering. The slope-coverage behavior of Three-Layer Calibration is therefore not an unexplained simulation artifact; the per-coefficient df arithmetic provides its dominant mechanical explanation, and the mechanism operates for the slope coefficients that are of primary interest in meta-regression.

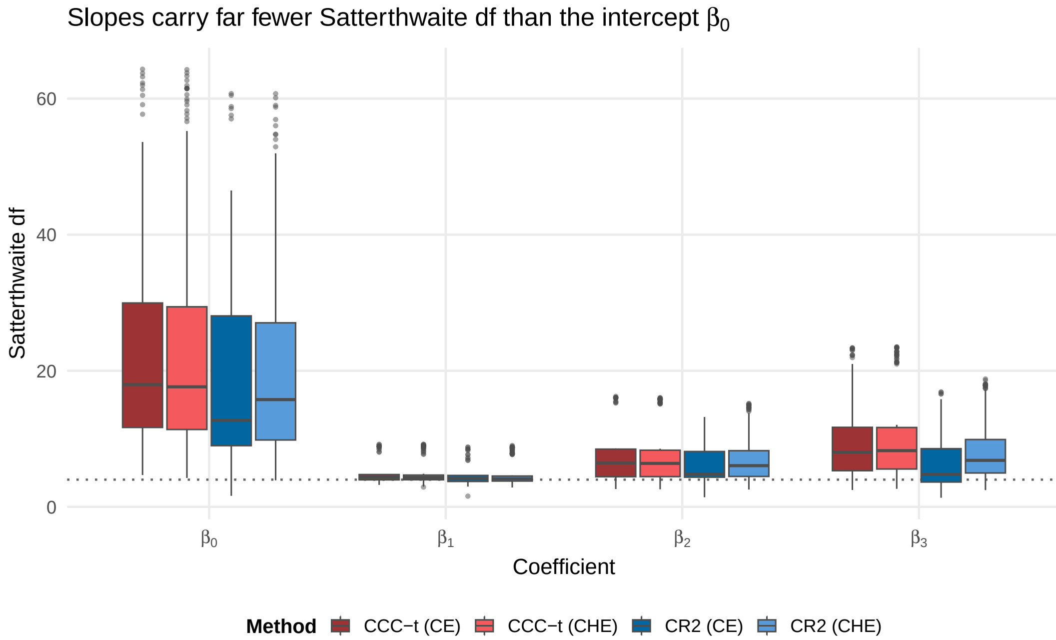

FIGURE 3. Per-condition Satterthwaite degrees of freedom by coefficient, for the finite-df methods on the same 177 correctly specified slope conditions. The average effect β_0 has a median df of 13.4, whereas the slope medians of 4.3–6.0 lie near the small-sample floor (dotted line at df = 4). Smaller df move the t quantile further from the normal quantile, so that CCC-t intervals widen in the settings where normal-reference intervals are too short; this df arithmetic is the dominant mechanical explanation for the CCC-t–CCC-q coverage gaps (0.070, 0.040, 0.037, 0.016), which are monotone in the df ordering.

4.4. **Costs and efficiency: power, width, and conservatism.** Calibration has costs, and on the small slope β_1 the coverage gain and the corresponding losses in power and width arise from a single mechanism. A slope about which the likelihood carries almost no information has a posterior that is wide, heavy-tailed, and shrunk toward zero; the near-floor df amplify that spread through the $t_{4.3}$ multiplier; and the result is a CCC-t interval whose median width is three orders of magnitude larger than that of CR2 (958–1,118 versus 0.95–0.99 across the CE and CHE arms). This interval covers at 0.978 and, for the same reason, rejects at 0.027 compared with 0.101 for CR2 (Figure 4); the high coverage and the low power are consequences of the same interval width. The cost, however, remains confined to the coefficient on which it is incurred. On the null slope β_3, CCC-t controls the Type-I error at 0.034 and CR2 at 0.045 (both acceptable), whereas CCC-q inflates to 0.070, which is the testing counterpart

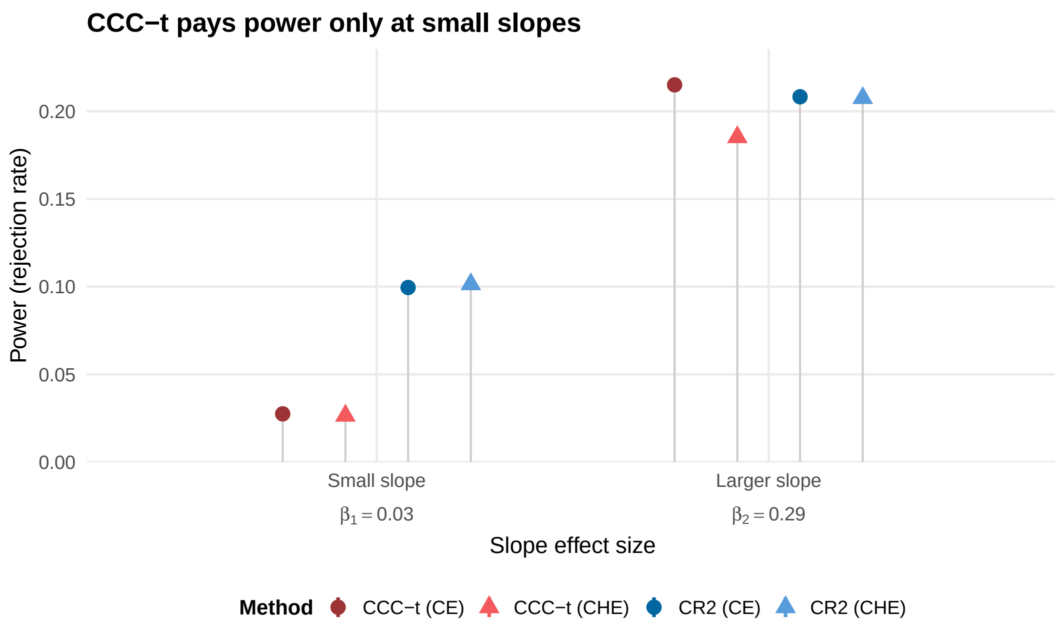

FIGURE 4. Power (rejection rate for true nonzero slopes) by effect-size class, over the 177 correctly specified slope conditions. At the small slope β_1 (true ≈ 0.03), CCC-t has lower power than CR2 (0.027 versus 0.101); this deficit is the cost of calibrated tails and shrinkage, and the 0.978 coverage of CCC-t on this coefficient is achieved by intervals whose median width is roughly 10^3 times that of CR2 (958–1,118 versus 0.95–0.99). At the larger slope β_2 (true ≈ 0.29) the methods are competitive (0.195 versus 0.208). Bars span ± 1.96 pooled binomial MCSEs.

of its undercoverage. On the substantial slope β_2, power is competitive (0.195 for CCC-t versus 0.208 for CR2), and CCC-t also produces narrower intervals on this coefficient (median width 1.11 versus 1.35) and a 21% smaller RMSE in the CE arm (relative RMSE 0.79; 0.90 in the CHE arm), which reflects the benefit of shrinkage in the Bayesian point estimate. The power deficit on β_1 is a small-effect phenomenon rather than a general loss: it arises because the procedure does not reject on the basis of evidence that the design barely contains. For analysts whose primary goal is the detection of small slopes, Figure 4 therefore constitutes an important caution.

The calibration of the standard errors completes the assessment of costs. On the scale-safe coefficient β_2, the median relative error of the model-based standard error with respect to the empirical one is +6.7% (CE arm) and +8.3% (CHE arm) for CCC-t (mild conservatism), compared with −4.9% and −1.6% for CR2 (mild anticonservatism). For β_0 the CCC-t figures are +0.7% and +0.6%. We note one reporting convention that was adopted throughout and fixed as a design decision of the simulation before any method comparison was run. Ratio-based metrics (relative SE error, calibration ratios) degenerate for coefficients whose true value is near zero; for β_1 the relative SE error reaches the order of $10^6\%$ purely because of the scale of the coefficient. Ratio metrics are therefore confined to β_0 and β_2 on mathematical grounds, whereas interval-scale metrics (coverage, width, and rejection rates) are reported for all coefficients (a full demonstration is given in Appendix E).

4.5. **Robustness and operational behavior.** The sensitivity blocks examine behavior at the boundaries of the design. Under contaminated (heavy-tailed) random effects, extreme within-study heterogeneity ($\omega = 0.3$), narrow and diffuse priors, and an alternative working correlation ($\rho_{\text{work}} = 0.6$), the qualitative ordering of Table 2 is unchanged (Appendix E; the prior and working-correlation blocks are intercept-only by design, a scope limitation that we note explicitly). Within the operating envelope the Bayesian arms converged in effectively all replications (median convergence rate 1.000; $\hat{R}$ and divergence diagnostics are given in Appendix E), and the worst-case sensitivity analysis, in which every refused or non-converged replication is counted as a failure to cover, does not alter the slope conclusions. The computational cost of CCC-t is essentially the cost of its MCMC fit (median 9.6–19.3 seconds per replication here, compared with 0.26 seconds for CR2); the calibration layers add no material cost.

5. Empirical illustration: brief alcohol interventions

We illustrate the workflow on the meta-analysis that anchored the simulation, namely the Tanner-Smith and Lipsey (2015) synthesis of brief alcohol interventions for adolescents and young adults, in the analytic form curated by Pustejovsky and Tipton (2022). The data comprise $J = 117$ studies contributing $N = 1{,}198$ standardized mean differences (Hedges' g), with a median of six effect sizes per study (range 1 to 108), across six outcome categories (frequency of use, frequency of heavy use, quantity of use, peak consumption, blood alcohol concentration, and combined measures). Within-study dependence is the defining feature of these data, since the trials measured multiple drinking outcomes at multiple follow-ups.

We fitted the cell-means meta-regression of effect size on outcome category plus five continuous moderators (centered follow-up week, long follow-up indicator, percent college, percent male, and attrition) under both the CE and the CHE working models, with $\beta_k \sim \mathrm{N}(0, 10^2)$, half-Cauchy$(0, 0.5)$ priors on τ and ω, four chains of

TABLE 3. Brief alcohol interventions (CE working model): outcome-category mean effects under the frequentist standard of care (`metafor` + `clubSandwich` CR2) and Three-Layer Calibration (`bayesRVE`). r_k is the per-coefficient calibration ratio (calibrated SD over raw posterior SD).

	CR2 (frequentist)			Three-Layer Calibration (Bayesian)			
Outcome category	Est.	SE	df	Est.	Calib. SD	df	r_k
Frequency of use	0.130	0.026	75.5	0.134	0.027	76.7	0.989
Frequency of heavy use	0.165	0.028	74.2	0.171	0.028	76.5	1.073
Quantity of use	0.153	0.024	81.6	0.159	0.024	93.3	0.955
Peak consumption	0.150	0.031	51.4	0.154	0.030	35.9	1.062
Blood alcohol concentration	0.180	0.028	61.1	0.184	0.028	47.2	0.941
Combined measures	0.155	0.032	40.9	0.159	0.032	28.5	1.007

Note. Estimates are standardized mean differences (Hedges' g), adjusted for the five continuous moderators. Frequentist column: REML with imputed $\rho = 0.6$ covariances. Bayesian column: posterior means and calibrated standard deviations; df are per-coefficient Satterthwaite values at posterior-mean plug-ins.

2,000 warmup and 2,000 sampling iterations ($S = 8{,}000$), and a fixed seed. Convergence was satisfactory ($\hat{R}_{\max} = 1.008$ for CE and 1.005 for CHE, with zero divergent transitions). The frequentist benchmark is the same working model fitted in `metafor`, with sampling covariances imputed under equicorrelation $\rho = 0.6$, and `clubSandwich` CR2 standard errors with Satterthwaite df. Table 3 reports the CE comparison for the six outcome-category means; the CHE table, the control-condition model, and full diagnostics are given in Appendix F.

Three observations follow from Table 3. First, the point estimates differ very little (differences of 0.003–0.013 SMD units). Layer 1 retains the Bayesian posterior means, and with $J = 117$ these lie close to the REML estimates, as theory predicts. Second, the calibrated standard deviations are close to the CR2 standard errors. The per-coefficient ratio of calibrated SD to CR2 SE averages 0.998 under CE (range 0.979–1.028) and 0.995 under CHE (range 0.925–1.054). Proposition 3.2 operates internally, in the sense that the calibrated draws match the `bayesRVE` target exactly by construction. These ratios therefore constitute a cross-implementation check. They compare the target-driven calibrated SD with an independent REML-based `clubSandwich` pipeline and show that the two agree closely despite the plug-in difference between posterior-mean and REML variance components. The pre-registered acceptance band for this validation (r within $[0.9, 1.1]$ for every coefficient) is satisfied for all twelve coefficients across both working models. Third, the calibration is not vacuous. The raw-posterior-to-target ratios r_k range from 0.94 to 1.08 across coefficients; the working-model posterior is too wide for some category means and too narrow for others, so that no global inflation factor could correct both directions at once. The matrix map (3.3) adjusts each coordinate as needed. The Satterthwaite df differ between the implementations (e.g., 35.9 versus 51.4 for peak consumption) because `bayesRVE` evaluates them at posterior-mean variance components whereas `metafor` uses REML values. When identical plug-in values are supplied, the two implementations agree to 10^{-6}, and at df > 30 the practical effect on the intervals is negligible. One feature of the output remains genuinely Bayesian after calibration and is worth noting. The posterior mean of τ (0.223 CE, 0.220 CHE) exceeds the REML estimate (0.141, 0.123), which is the familiar consequence of a heavy-tailed prior on a variance component. Layer 2 calibrates the spread of $\boldsymbol{\beta}$, not the analyst's beliefs about heterogeneity, and the two coexist coherently in the calibrated report.

6. Discussion

Bayesian meta-analysis and robust variance estimation have developed as parallel solutions to the same problem, and the analyst has been asked to choose between the estimation advantages of the former and the inferential guarantees of the latter. Three-Layer Calibration reduces this tradeoff: the Bayesian point estimate and heterogeneity report can be paired with an uncertainty target calibrated to the RVE standard of care. The posterior mean is retained (Layer 1); the posterior spread is transformed, exactly and by an algebraic identity, onto the CR2 cluster-sandwich covariance that frequentist research synthesis has established as the appropriate uncertainty target for dependent effect sizes (Layer 2); and intervals are referred to per-coefficient Satterthwaite t distributions rather than to a normal distribution (Layer 3). In 324 simulation conditions built on a real meta-analytic database, the resulting intervals attained coverage of 0.952 for the average effect and coverage within the range 0.966–0.978 across the moderator slopes. The normal-reference variant (CCC-q) undercovered all three slopes (0.908–0.931). The frequentist standard of care was at or above the nominal level on two slopes (0.955 and 0.961) but fell to 0.905 on the small slope, the setting in which the calibrated interval achieves its largest gain and also incurs its power and width costs (Section 4.4). The advantage was attributable to the slopes' roughly threefold degrees-of-freedom deficit and increased monotonically with that deficit. On real data, the calibrated Bayesian standard deviations matched the frequentist CR2 standard errors to within one percent on average, with no single coefficient deviating by more than eight percent, while Bayesian point estimation and heterogeneity inference were retained.

Our recommendations for practice are conservative. We recommend choosing the working model from the structure of the data, following the decision framework of Pustejovsky and Tipton (2022); in our simulations the calibrated intervals behaved similarly under the two working models, and fitting the CHE model when the data were generated under the simpler CE structure cost little (coverage of 0.977 on the small slope under overfitting, Figure 2). We further recommend calibrating to the CR2 target with Satterthwaite tails (the package defaults), respecting the operating envelope $J \geq \max(8, 2K + 2)$, and reporting the calibration diagnostics r_k alongside the intervals, since these diagnostics quantify, coefficient by coefficient, the extent to which the working-model posterior required correction, which is itself scientific information about the dependence structure. On the substantial slope β_2, calibration carried no efficiency penalty relative to CR2: the intervals were narrower, power was competitive, and the point estimate had a smaller root mean squared error (Section 4.4). When a specific small slope is the scientific target and its detection is the primary goal, the power cost documented in Section 4.4 should be taken into account, and a frequentist CR2 analysis (or a design with more studies) may be preferable; the two analyses are inexpensive to run side by side.

The approach has several limitations. The exact-matching propositions guarantee covariance matching but not coverage; coverage was demonstrated under data-generating processes anchored to one large empirical database, with equicorrelated-plus-hierarchical dependence and standardized mean differences, and we make no claims beyond that envelope. The CR2 target is constructed at plug-in variance components, so a poorly estimated ω^2 shifts the target itself; the working correlation enters the frequentist comparator and the sensitivity analyses, and non-exchangeable within-study structures (autoregressive follow-ups, unstructured multivariate outcomes) are outside the current implementation. Calibration is carried out per coefficient; multi-parameter Wald-type tests require a joint calibrated reference and are a subject for future development rather than a current feature. The calibrated intervals are mildly conservative in standard-error terms on the scale-safe nonzero slope β_2 (+7–8%; ratio metrics are not meaningfully defined for the near-zero slopes) and uniformly conservative in coverage terms on all three slopes (0.966–0.978 against a nominal 0.95); this conservatism is in the safe direction but is nevertheless a bias. Finally, the procedure inherits the computational cost of MCMC (seconds to minutes per fit, compared with fractions of a second for CR2), which is relevant for large-scale reanalyses.

We briefly note two extensions. First, the per-coefficient calibration ratios r_k are an instance of a broader system of design-effect diagnostics, namely ratios of robust to model-based uncertainty computed coefficient by coefficient, developed to date for Bayesian survey inference (Lee, 2026). A meta-analytic elaboration of these diagnostics, which would indicate which coefficients need calibration before any correction is applied, is companion work rather than a claim of the present article. Second, the same three-layer logic (a model-based location, a named robust target, and small-sample tails) applies whenever a Bayesian model is fitted under working assumptions about clustered dependence that the analyst does not fully believe; network meta-analysis and multivariate synthesis are natural next applications. We conclude that the choice between Bayesian modeling and robust inference is not forced, because the sandwich covariance matrix provides an explicit target and an affine transformation maps the posterior draws onto that target exactly.

Software and reproducibility. Three-Layer Calibration is implemented in the open-source R package `bayesRVE` (Lee & Tipton, 2026) (https://github.com/joonho112/bayesRVE; documentation at https://joonho112.github.io/bayesRVE/), which draws posterior samples through Stan (Carpenter et al., 2017). A replication package containing the simulation code, the full condition grid, and all result files reproducing every number, table, and figure in this article is available at https://github.com/joonho112/bayesRVE-replication (companion guide at https://joonho112.github.io/bayesRVE-replication/). The merged replication-level simulation results (the analytic bulk) are archived on Zenodo at https://doi.org/10.5281/zenodo.21259722. Complete documentation of the simulation design, verification, and full results is available at https://joonho112.github.io/bayesRVE-simulation-study/.

Data availability. The empirical data analyzed in Section 5 are from the published meta-analysis of Tanner-Smith and Lipsey (2015), as curated in the publicly available replication materials of Pustejovsky and Tipton (2022). The data belong to the original authors and to those materials and are not redistributed with our code.

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Online Supplement

Three-Layer Calibration for Bayesian Robust Variance Estimation in Meta-Analysis With Dependent Effect Sizes

JoonHo Lee Elizabeth Tipton

This supplement accompanies the main text. In Appendix A we collect notation and the estimator and metric catalogs. In Appendix B we derive the CR2 cluster-sandwich calibration target. In Appendix C we state and prove the exact-calibration propositions and document the numerical verification protocol. In Appendix D we describe the per-coefficient Satterthwaite construction and its parity with `clubSandwich`. Appendix E reports the complete simulation protocol and full results, and Appendix F the complete empirical results. Finally, Appendix G documents software and reproducibility. Section, equation, table, and figure numbers below are prefixed by the appendix letter; references to "the main text" cite the manuscript by section number.

Contents

Appendix A. Notation, estimator catalog, and metric catalog

A.1. **Notation.** Table A.1 fixes the notation used in the main text and throughout this supplement. The following conventions apply. Studies are indexed by j and effect sizes within a study by i; bold lowercase symbols denote vectors and bold uppercase symbols denote matrices; hats denote posterior means (for Bayesian objects) or plug-in estimates; and starred draws are calibrated draws.

TABLE A.1. Core notation.

Symbol	Meaning
J	number of studies (independent clusters)
n_j	number of effect sizes in study j; $N = \sum_j n_j$
y_{ij}, $\mathbf{y}_j$	effect size estimate i in study j; stacked n_j-vector
v_{ij}, $\bar{s}_j^2$	known sampling variance; study mean $n_j^{-1} \sum_i v_{ij}$
$\mathbf{x}_{ij}$, $\mathbf{X}_j$, $\bar{\mathbf{x}}_j$	moderator vector ($K \times 1$); study design matrix ($n_j \times K$); column mean
$\boldsymbol{\beta}$	meta-regression coefficients ($K \times 1$)
θ_j, τ^2	study random effect; between-study variance
w_{ij}, ω^2	effect-level random effect (CHE); within-study variance
ρ	within-study sampling correlation (conditional, among sampling errors)
a_j	$\bar{s}_j^2$ (CE) or $\bar{s}_j^2 + \hat{\omega}^2$ (CHE)
$\boldsymbol{\Sigma}_j$	working covariance $a_j \mathbf{I}_{n_j} + \tau^2 \mathbf{1}\mathbf{1}^\intercal$
v_j	study-mean marginal variance $a_j/n_j + \tau^2$
$\mathbf{W}_j$	working weights $\boldsymbol{\Sigma}_j^{-1}$
$\mathbf{Q}$	WLS bread $(\sum_j \mathbf{X}_j^\intercal \mathbf{W}_j \mathbf{X}_j)^{-1}$
$\hat{\mathbf{e}}_j$	working-model residuals $\mathbf{y}_j - \mathbf{X}_j\hat{\boldsymbol{\beta}}$ at the posterior mean
$\mathbf{A}_j$	CR2 adjustment matrix of study j
$\mathbf{V}_{\text{CR2}}$	CR2 cluster-sandwich target for $\hat{\boldsymbol{\beta}}$
$\boldsymbol{\psi}^{(s)}$, $\hat{\boldsymbol{\psi}}$, $\boldsymbol{\psi}^{*(s)}$	MCMC draw $s = 1, \ldots, S$; posterior mean; calibrated draw
$\hat{\boldsymbol{\Sigma}}_{\text{MCMC}}$	sample covariance of MCMC draws (block)
$\mathbf{L}_1$, $\mathbf{L}_2$	lower Cholesky factors of $\hat{\boldsymbol{\Sigma}}_{\text{MCMC}}$ and $\mathbf{V}_{\text{target}}$
$\mathbf{A}_{\text{ccc}} = \mathbf{L}_2\mathbf{L}_1^{-1}$	CCC transformation matrix
r_k	per-coefficient calibration ratio $\{[\mathbf{V}_{\text{target}}]_{kk}/[\hat{\boldsymbol{\Sigma}}_{\text{MCMC}}]_{kk}\}^{1/2}$
df_k	per-coefficient Satterthwaite degrees of freedom

Two points of terminology should be noted. First, ρ always denotes the conditional correlation among within-study sampling errors, not the marginal correlation between effect sizes; the latter is inflated by the shared random effects, since the marginal covariance adds τ^2, and under CHE ω^2, to every within-study pair. Second, the center preserved by calibration is the posterior mean, that is, the Monte Carlo average of the draws, and not the posterior mode.

A.2. **Estimator catalog (M1–M9).** The simulation applies nine estimators to every generated dataset. Table A.2 gives the complete definitions, and the main text groups the estimators into five families. Estimators M1/M3/M7 share one Stan fit, as do M2/M6/M8, so all Bayesian contrasts are within-fit; differences among CCC-t, CCC-q, and uncalibrated Bayes therefore reflect the inference layers only and not MCMC noise.

TABLE A.2. Estimator catalog. "Working model" is the fitted structure; crossing CE/CHE estimators with CE/CHE generating structures defines the specification roles (correct, overfit, misspecified).

ID	Label	Definition
M1	CCC-t (CE)	Bayesian CE fit; CR2 sandwich target at posterior-mean plug-ins; blockwise CCC; per-coefficient Satterthwaite t intervals. The proposed method, CE arm.
M2	CCC-t (CHE)	As M1 with the CHE working model (ω estimated; $\boldsymbol{\beta}$-block calibration only, the study effects being integrated out). The proposed method, CHE arm.
M3	Uncalibrated Bayes (CE)	Posterior 2.5%/97.5% credible interval from the M1 fit; no calibration layers.
M4	CR2 (CE)	`metafor::rma.mv` (Viechtbauer, 2010) CE fit with sampling covariances imputed at ρ_{work}; `clubSandwich` (Pustejovsky, 2026) CR2 standard errors with Satterthwaite t. Frequentist benchmark, CE arm.
M5	CR2 (CHE)	As M4 with the two-level (CHE) structure. Frequentist benchmark, CHE arm.
M6	Uncalibrated Bayes (CHE)	Posterior credible interval from the M2 fit.
M7	CCC-q (CE)	Identical calibrated draws as M1; interval endpoints from the calibrated draws under a normal reference (df $= \infty$). Isolates Layer 3 (identical draws, normal reference), CE arm.
M8	CCC-q (CHE)	Isolates Layer 3, CHE arm (shares the M2 fit).
M9	`robumeta` (CE)	`robumeta::robu` (Fisher et al., 2023) with correlated-effects weights ($\rho = \rho_{\text{work}}$) and the small-sample correction; applicable to the intercept-only design ($K = 1$, CE) only. External RVE benchmark.

A.3. **Metric catalog.** For each condition $\times$ method $\times$ coefficient cell, the pipeline computes the following quantities: bias $\bar{\hat{\beta}} - \beta_{\text{true}}$; empirical SE (the SD of $\hat{\beta}$ across replications); model SE $\{\text{mean}(\text{SE}^2)\}^{1/2}$; relative SE error $100(\text{ModSE}/\text{EmpSE}-1)\%$; coverage of nominal 95% intervals (the primary metric, computed over converged replications only, with a count-as-miss sensitivity analysis in Appendix E.4); bias-eliminated coverage; mean and median interval width; RMSE and RMSE relative to the same-arm CR2 reference; relative precision; rejection rate (power for $\beta_{\text{true}} \neq 0$ and Type-I error for $\beta_3 = 0$); the calibration diagnostics r_k, $\|\mathbf{A}_{\text{ccc}} - \mathbf{I}\|_F$, and the CCC skip rate (for CCC methods only); convergence rate (with all attempted replications as the denominator); the mean Satterthwaite df (for finite-df methods); and wall-clock time per replication. Every point estimate is stored with a Monte Carlo standard error. Ratio-based metrics are subject to the scale-degeneracy policy of Appendix E.10.

Appendix B. The CR2 cluster-sandwich calibration target

This appendix derives the calibration target used by Layer 2, namely the bias-reduced CR2 cluster sandwich for the fixed-effects block and the per-study conditional variances for the study-effects block. All objects are evaluated at plug-in variance components and at the posterior mean $\hat{\boldsymbol{\beta}}$. The plug-in variance components are the posterior means $\hat{\tau}^2$ and, for CHE, the Jensen-consistent $\hat{\omega}^2 = S^{-1}\sum_s(\omega^{(s)})^2$. The latter is the posterior mean of the variance, which respects $\mathbb{E}[\omega^2] \geq (\mathbb{E}[\omega])^2$.

B.1. **Why the full-joint sandwich cannot be the target.** Write the CE conditional parameter vector as $\boldsymbol{\psi} = (\boldsymbol{\beta}^\intercal, \theta_1, \ldots, \theta_J)^\intercal$, of dimension $d = K + J$. A sandwich for the full vector would use the meat $\mathbf{M}_{\text{full}} = \sum_{j=1}^J \mathbf{s}_j\mathbf{s}_j^\intercal$, where $\mathbf{s}_j$ is study j's d-dimensional score contribution. Because $\mathbf{M}_{\text{full}}$ is a sum of J rank-one matrices, we have

$$\operatorname{rank}(\mathbf{M}_{\text{full}}) \;\leq\; J \;<\; K + J = d, \tag{B.1}$$

so the full-joint meat is structurally singular whenever $K \geq 1$. That is, there are only J independent clusters from which to estimate the $(K + J)$-dimensional between-cluster variability. This is the incidental-parameters geometry of Neyman and Scott (1948), in which each θ_j appears in exactly one cluster's score. A singular target admits no Cholesky factor, so full-joint calibration is undefined rather than merely inadvisable. Furthermore, in the conditional parameterization the θ_j absorb nearly all between-study variation, which drives the $\boldsymbol{\beta}$-block of $\mathbf{M}_{\text{full}}$ toward zero and the implied variance far below the true sampling variance of $\hat{\boldsymbol{\beta}}$.

The calibration therefore proceeds blockwise, with a full $K \times K$ marginal target for $\boldsymbol{\beta}$ (this appendix) and per-study scalar conditional targets for the θ_j (Appendix B.5). The marginal fixed-effects target is chosen for two independent reasons. First, the rank obstruction above disqualifies the full-joint object. Second, the reporting estimand of a meta-regression is the population-level $\boldsymbol{\beta}$, which integrates over study effects, so the marginal sandwich is the variance of the quantity that is actually reported. A population-level treatment of the rank obstruction and of the estimand-based selection of the marginal fixed-effects target is given in the companion article (Lee et al., 2026).

B.2. **Working covariance, weights, and the Woodbury forms.** Under the working models of the main text (Section 2.1), study j's working covariance is compound symmetric,

$$\boldsymbol{\Sigma}_j \;=\; a_j \mathbf{I}_{n_j} + \tau^2 \mathbf{J}_{n_j}, \qquad a_j = \begin{cases} \bar{s}_j^2 & \text{(CE)} \\ \bar{s}_j^2 + \hat{\omega}^2 & \text{(CHE)}, \end{cases} \tag{B.2}$$

with $\mathbf{J}_{n_j} = \mathbf{1}\mathbf{1}^\intercal$. Its two distinct eigenvalues are $c_j = a_j + n_j\tau^2$ (eigenvector $\propto \mathbf{1}$) and a_j (multiplicity $n_j - 1$), giving the Woodbury inverse and the closed-form symmetric square roots

$$\mathbf{W}_j = \boldsymbol{\Sigma}_j^{-1} = \frac{1}{a_j}\mathbf{I}_{n_j} - \frac{\tau^2}{a_j\, c_j}\mathbf{J}_{n_j}, \tag{B.3}$$

$$\mathbf{W}_j^{1/2} = \frac{1}{\sqrt{a_j}}\mathbf{I}_{n_j} + \frac{1}{n_j}\Big(\frac{1}{\sqrt{c_j}} - \frac{1}{\sqrt{a_j}}\Big)\mathbf{J}_{n_j}, \tag{B.4}$$

$$\mathbf{W}_j^{-1/2} = \sqrt{a_j}\,\mathbf{I}_{n_j} + \frac{1}{n_j}\big(\sqrt{c_j} - \sqrt{a_j}\big)\mathbf{J}_{n_j}. \tag{B.5}$$

These closed forms reduce every $n_j \times n_j$ inversion or square root in the pipeline from $O(n_j^3)$ to $O(n_j)$ array operations. This reduction makes the CR2 adjustment matrices computationally affordable inside a simulation comprising hundreds of thousands of fits. Setting $\hat{\omega}^2 = 0$ recovers the CE forms, and an eigendecomposition fallback guards against the numerical edge case of non-positive computed eigenvalues. Averaging yields the study-mean marginal variance $\mathrm{Var}(\bar{y}_j) = a_j/n_j + \tau^2 \equiv v_j$.

B.3. **Bread: the weighted-least-squares matrix.** The bread of the target is the WLS matrix

$$\mathbf{Q} \;=\; \Big(\sum_{j=1}^{J} \mathbf{X}_j^\intercal \mathbf{W}_j \mathbf{X}_j\Big)^{-1}, \tag{B.6}$$

not the posterior Hessian. The distinction matters for two reasons. First, CR2 is defined through the WLS hat matrix, and replacing $\mathbf{Q}$ by a prior-curved Hessian would change the leverage decomposition on which the exactness of the correction rests. Second, using $\mathbf{Q}$ keeps the target identical (given identical plug-ins) to the frequentist standard of care. This is the purpose of the calibration: the prior is allowed to influence the location of the posterior (Layer 1) but not the spread that is treated as calibrated (Layer 2). When all moderators are constant within studies, (B.6) collapses to the study-level form $\mathbf{Q} = (\sum_j \bar{\mathbf{x}}_j \bar{\mathbf{x}}_j^\intercal / v_j)^{-1}$. The implementation supports moderators that vary within studies through the general form.

B.4. **Residual contraction and the CR2 adjustment.** Let $\hat{\mathbf{e}}_j = \mathbf{y}_j - \mathbf{X}_j\hat{\boldsymbol{\beta}}$ and define the cross-study hat blocks $\mathbf{H}_{jl} = \mathbf{X}_j\mathbf{Q}\mathbf{X}_l^\intercal\mathbf{W}_l$. Under the generating model $\mathbf{y}_l = \mathbf{X}_l\boldsymbol{\beta} + \boldsymbol{\varepsilon}_l$ with $\mathrm{Var}(\boldsymbol{\varepsilon}_l) = \boldsymbol{\Sigma}_l^{\mathrm{true}}$,

$$\begin{aligned}\hat{\mathbf{e}}_j &= (\mathbf{I} - \mathbf{H}_{jj})\boldsymbol{\varepsilon}_j - \sum_{l\neq j}\mathbf{H}_{jl}\boldsymbol{\varepsilon}_l,\\ \mathbb{E}\big[\hat{\mathbf{e}}_j\hat{\mathbf{e}}_j^\intercal\big] &= (\mathbf{I} - \mathbf{H}_{jj})\boldsymbol{\Sigma}_j^{\mathrm{true}}(\mathbf{I} - \mathbf{H}_{jj})^\intercal + \sum_{l\neq j}\mathbf{H}_{jl}\boldsymbol{\Sigma}_l^{\mathrm{true}}\mathbf{H}_{jl}^\intercal. \end{aligned} \tag{B.7}$$

The leading term is a strict contraction of $\boldsymbol{\Sigma}_j^{\mathrm{true}}$, because the fitted model absorbs part of each study's error. Substituting raw residuals into the sandwich meat therefore biases it downward, and the deficit grows with study leverage. This deficit is the small-J anticonservatism of uncorrected RVE. The CR2 correction (Bell & McCaffrey, 2002; Tipton, 2015) specifies adjustment matrices $\mathbf{A}_j$ satisfying

$$\mathbf{A}_j(\mathbf{I} - \mathbf{H}_{jj})\boldsymbol{\Sigma}_j(\mathbf{I} - \mathbf{H}_{jj})^\intercal\mathbf{A}_j^\intercal \;=\; \boldsymbol{\Sigma}_j, \tag{B.8}$$

so that, under the working model and up to the $O(1/J)$ cross-study term in (B.7), the adjusted meat is unbiased. Conjugating by $\mathbf{W}_j^{1/2}$ turns (B.8) into a symmetric problem with solution

$$\mathbf{A}_j \;=\; \mathbf{W}_j^{-1/2}\big(\mathbf{I}_{n_j} - \tilde{\mathbf{H}}_{jj}\big)^{-1/2}\mathbf{W}_j^{1/2}, \qquad \tilde{\mathbf{H}}_{jj} = \mathbf{W}_j^{1/2}\mathbf{X}_j\mathbf{Q}\mathbf{X}_j^\intercal\mathbf{W}_j^{1/2}, \tag{B.9}$$

where $\tilde{\mathbf{H}}_{jj}$ is symmetric by construction and shares eigenvalues with $\mathbf{H}_{jj}$. The implementation computes $(\mathbf{I} - \tilde{\mathbf{H}}_{jj})^{-1/2}$ by eigendecomposition with eigenvalues truncated below 10^{-10}, and issues a leverage warning when $\max_j h_j > 0.5$ (with $h_j = \bar{\mathbf{x}}_j^\intercal\mathbf{Q}\bar{\mathbf{x}}_j/v_j$ in the study-level path). In the scalar case ($n_j = 1$) the adjustment reduces to the familiar $(1 - h_j)^{-1/2}$ of HC2-type corrections, and with uniform leverage and $n_j \equiv 1$ the CR2 sandwich recovers the classical $J/(J-K)$ degrees-of-freedom correction.

The target is assembled from adjusted scores:

$$\tilde{\mathbf{s}}_j = \mathbf{X}_j^\intercal\mathbf{W}_j\mathbf{A}_j\hat{\mathbf{e}}_j, \qquad\qquad \mathbf{V}_{\mathrm{CR2}} \;=\; \mathbf{Q}\Big(\sum_{j=1}^{J}\tilde{\mathbf{s}}_j\tilde{\mathbf{s}}_j^\intercal\Big)\mathbf{Q}. \tag{B.10}$$

A scalar global correction (e.g., $J/(J-K)\cdot\mathbf{V}_{\mathrm{CR0}}$) cannot substitute for (B.10), since the bias of the raw meat is study-specific, matrix-valued, and concentrated in high-leverage clusters. Moreover, the empirical calibration ratios reported in Table 3 of the main text show corrections in both directions across coefficients within a single dataset.

We conclude this subsection with two remarks on the scope of the target. First, the meat in (B.10) accumulates likelihood scores only. Prior scores are excluded because the purpose of the meat is to measure data-driven between-cluster variability, and the prior's $O(1/J)$ influence on the location is already carried by $\hat{\boldsymbol{\beta}}$ and the residuals. Second, the entire target is conditional on the plug-in variance components, so the robustness of (B.10) is to misspecification of the within-study dependence, not to gross error in $\hat{\omega}^2$ itself, which would shift the weights, the bread, and hence the target.

B.5. **The study-effects block.** For the CE model the calibrated report also covers the study effects. Conditional on $\hat{\boldsymbol{\beta}}$ and $\hat{\tau}^2$, study j's effect θ_j has a scalar conditional score and Hessian. Because the θ_j are conditionally independent given $(\boldsymbol{\beta}, \tau^2)$, the resulting conditional sandwich variances $V_{jj}^{(\theta)}$ are collected in a diagonal target $\mathbf{V}^{(\theta)}$. The theta-block calibration in Appendix C.3 therefore reduces to J scalar rescalings, and off-diagonal posterior correlations among the θ_j are transformed but not targeted. This is deliberate, since the conditional target itself carries no cross-study covariance. The CHE model integrates the θ_j out of the likelihood, so there is no theta block and only $\boldsymbol{\beta}$ is calibrated.

Appendix C. Cholesky Calibration Correction: proofs and verification

C.1. **The transformation.** Let $\boldsymbol{\psi}^{(1)}, \ldots, \boldsymbol{\psi}^{(S)}$ be MCMC draws of a parameter block of dimension d, with sample mean $\hat{\boldsymbol{\psi}} = S^{-1}\sum_s \boldsymbol{\psi}^{(s)}$ and sample covariance $\hat{\boldsymbol{\Sigma}}_{\text{MCMC}} = (S-1)^{-1}\sum_s(\boldsymbol{\psi}^{(s)} - \hat{\boldsymbol{\psi}})(\boldsymbol{\psi}^{(s)} - \hat{\boldsymbol{\psi}})^{\intercal}$, assumed positive definite (which requires $S > d$ and non-degenerate draws). Let $\mathbf{V}_{\text{target}}$ be a positive-definite target. With lower-triangular Cholesky factors $\mathbf{L}_1 = \text{chol}(\hat{\boldsymbol{\Sigma}}_{\text{MCMC}})$ and $\mathbf{L}_2 = \text{chol}(\mathbf{V}_{\text{target}})$, the Cholesky Calibration Correction maps every draw by

$$\boldsymbol{\psi}^{*(s)} \;=\; \hat{\boldsymbol{\psi}} + \mathbf{A}_{\text{ccc}}\big(\boldsymbol{\psi}^{(s)} - \hat{\boldsymbol{\psi}}\big), \qquad \mathbf{A}_{\text{ccc}} \;=\; \mathbf{L}_2\mathbf{L}_1^{-1}. \tag{C.1}$$

In the scalar case, (C.1) reduces to $\psi^{*(s)} = \hat{\psi} + r\,(\psi^{(s)} - \hat{\psi})$ with $r = (V_{\text{target}}/\hat{\Sigma}_{\text{MCMC}})^{1/2}$. The dispersion of the draws is therefore increased when the posterior is narrower than the target ($r > 1$) and reduced when the posterior is wider than the target ($r < 1$).

C.2. **Exactness propositions.**

Proposition C.1 (Center preservation). $\bar{\boldsymbol{\psi}}^* \equiv S^{-1}\sum_{s=1}^{S}\boldsymbol{\psi}^{*(s)} = \hat{\boldsymbol{\psi}}$, *exactly, for any matrix* $\mathbf{A}_{\text{ccc}}$.

Proof. By linearity of the sample mean, $\bar{\boldsymbol{\psi}}^* = \hat{\boldsymbol{\psi}} + \mathbf{A}_{\text{ccc}}\big(S^{-1}\sum_s(\boldsymbol{\psi}^{(s)} - \hat{\boldsymbol{\psi}})\big) = \hat{\boldsymbol{\psi}} + \mathbf{A}_{\text{ccc}} \cdot \mathbf{0} = \hat{\boldsymbol{\psi}}$, where the inner sum vanishes because $\hat{\boldsymbol{\psi}}$ is defined as the sample mean of the draws. Note that the fixed point of the affine map is the posterior mean; if the transformation were centered at any other location (e.g., the mode), this cancellation would not occur. □

Proposition C.2 (Exact covariance matching)**.** *The sample covariance of the calibrated draws equals the target exactly:* $\mathrm{Cov}(\boldsymbol{\psi}^{*(1)}, \ldots, \boldsymbol{\psi}^{*(S)}) = \mathbf{V}_{\text{target}}$.

Proof. Write $\tilde{\boldsymbol{\psi}}^{(s)} = \boldsymbol{\psi}^{(s)} - \hat{\boldsymbol{\psi}}$. By Proposition C.1, $\boldsymbol{\psi}^{*(s)} - \bar{\boldsymbol{\psi}}^* = \mathbf{A}_{\text{ccc}}\tilde{\boldsymbol{\psi}}^{(s)}$, so

$$\begin{aligned}\mathrm{Cov}(\boldsymbol{\psi}^*) &= \frac{1}{S-1}\sum_{s=1}^{S} \mathbf{A}_{\text{ccc}}\tilde{\boldsymbol{\psi}}^{(s)}\tilde{\boldsymbol{\psi}}^{(s)\intercal}\mathbf{A}_{\text{ccc}}^{\intercal} = \mathbf{A}_{\text{ccc}}\,\hat{\boldsymbol{\Sigma}}_{\text{MCMC}}\,\mathbf{A}_{\text{ccc}}^{\intercal} \\ &= \mathbf{L}_2\mathbf{L}_1^{-1}\,\mathbf{L}_1\mathbf{L}_1^{\intercal}\,\mathbf{L}_1^{-\intercal}\mathbf{L}_2^{\intercal} = \mathbf{L}_2\mathbf{L}_2^{\intercal} = \mathbf{V}_{\text{target}}.\end{aligned}$$

Each equality is algebraic; no appeal is made to the distribution of the draws, to the sample size S (beyond positive definiteness of $\hat{\boldsymbol{\Sigma}}_{\text{MCMC}}$), to the correctness of the working model, or to any asymptotic argument. □

Proposition C.3 (Identity under exact covariance matching)**.** *If* $\hat{\boldsymbol{\Sigma}}_{\text{MCMC}} = \mathbf{V}_{\text{target}}$*, then* $\mathbf{A}_{\text{ccc}} = \mathbf{I}_d$ *and* $\boldsymbol{\psi}^{*(s)} = \boldsymbol{\psi}^{(s)}$ *for every* s*.*

Proof. The Cholesky factorization with positive diagonal is unique, so $\hat{\boldsymbol{\Sigma}}_{\text{MCMC}} = \mathbf{V}_{\text{target}}$ implies $\mathbf{L}_1 = \mathbf{L}_2$ and hence $\mathbf{A}_{\text{ccc}} = \mathbf{L}_2\mathbf{L}_1^{-1} = \mathbf{I}_d$. □

Proposition C.3 shows that the calibration returns the draws unchanged when the fitted posterior already has the target covariance, and this property justifies applying the transformation unconditionally rather than making its use conditional on a preliminary test. Note that correct specification of the working model does not by itself guarantee this equality in finite samples, because the CR2 small-sample adjustment and the plug-in variance components can leave a nontrivial map even under a correctly specified model; the argument for unconditional application therefore rests on the identity property rather than on the correctness of the working model. We emphasize the scope of these guarantees once more. The propositions establish covariance matching, which is an exact finite-sample property. Frequentist coverage of intervals constructed from the calibrated draws is a separate operating characteristic that is asymptotic in J; it additionally requires the CR2 target to estimate the true sampling variance well, and it is assessed empirically in Appendix E.

C.3. **Blockwise structure and the CHE dispatch.** For the CE model the calibrated vector is split into the $\boldsymbol{\beta}$ block and the $\boldsymbol{\theta}$ block, with block-diagonal target

$$\mathbf{V}_{\mathrm{CCC}} = \begin{pmatrix} \mathbf{V}_{\mathrm{CR2}} & \mathbf{0} \\ \mathbf{0} & \mathbf{V}^{(\theta)} \end{pmatrix}, \tag{C.2}$$

where the $\boldsymbol{\beta}$ block is calibrated by the full matrix map (C.1) and the $\boldsymbol{\theta}$ block, whose target is diagonal (Appendix B.5), is calibrated by J scalar rescalings $\theta_j^{*(s)} = \hat{\theta}_j + r_j(\theta_j^{(s)} - \hat{\theta}_j)$ with $r_j = (V_{jj}^{(\theta)}/\hat{\Sigma}_{\mathrm{MCMC},jj}^{(\theta)})^{1/2}$. Propositions C.1 and C.2 then hold within each block, so that $\mathrm{Cov}(\boldsymbol{\beta}^*) = \mathbf{V}_{\mathrm{CR2}}$ exactly and $\mathrm{Var}(\theta_j^*) = V_{jj}^{(\theta)}$ exactly for each j. The cross-block covariance $\mathrm{Cov}(\boldsymbol{\beta}^*, \boldsymbol{\theta}^*)$ is transformed but not targeted, and after the theta-block rescaling $\mathrm{Cov}(\theta_j^*, \theta_{j'}^*) = r_j r_{j'} \hat{\Sigma}_{\mathrm{MCMC},jj'}^{(\theta)}$. We do not regard either property as a deficiency, since no defensible cross-block or cross-study target exists (Appendix B.1). The rationale for the block split is given in Appendix B.1; the split also reduces the computational cost from $O((K+J)^3)$ to $O(K^3 + J)$.

For the CHE model the MCMC parameterization integrates the θ_j out, so only the $\boldsymbol{\beta}$ block exists; the implementation dispatches on the working model and suppresses theta output downstream. In either case the Layer-2 cost is $O(SK^2 + K^3)$, which is sub-millisecond at $S = 8{,}000$ and $K \leq 12$.

We now describe two implementation safeguards. The MCMC covariance receives a nearest-positive-definite repair only if numerical noise leaves it marginally non-PD; the target is never repaired, since a non-PD target signals a structural problem (too few clusters, extreme leverage) that a repair would mask. If the $\boldsymbol{\beta}$-block target fails positive definiteness, the calibration is skipped and flagged rather than carried out in a silently degraded form; skip rates in the simulation are reported in Appendix E.6 (they are zero for CCC-t).

C.4. **Numerical verification protocol.** Because the propositions are exact, their implementation can be tested to machine precision, and the package's automated suite performs the following checks:

- **V1 (center)**: the suite requires $\max_k |\bar{\psi}_k^* - \hat{\psi}_k| < 10^{-6}$ on every fit; the observed deviations are of order 10^{-14}–10^{-12} (IEEE 754 double precision).
- **V2 (diagonal matching)**: each ratio $\mathrm{Var}(\beta_k^*)/[\mathbf{V}_{\mathrm{CR2}}]_{kk}$ must lie within 5% of unity; the observed deviations are of order 10^{-13}.
- **V3 (identity)**: synthetic configurations with $\mathbf{V}_{\mathrm{target}} := \hat{\boldsymbol{\Sigma}}_{\mathrm{MCMC}}$ return $\|\mathbf{A}_{\mathrm{ccc}} - \mathbf{I}\|_F < 10^{-12}$.

- **V4 (full-matrix matching, $K \geq 2$)**: the relative Frobenius error $\|\mathrm{Cov}(\boldsymbol{\beta}^*) - \mathbf{V}_{\mathrm{CR2}}\|_F / \|\mathbf{V}_{\mathrm{CR2}}\|_F$ must be below 5%; the observed errors are at machine precision.

The 5% test thresholds are deliberately loose tolerances for automated testing; the observed deviations are those of floating-point arithmetic, as the propositions require.

Appendix D. Per-coefficient Satterthwaite degrees of freedom

D.1. **Construction.** Layer 3 replaces the normal reference distribution with a per-coefficient t reference whose degrees of freedom approximate the sampling variability of the CR2 variance estimate itself, following the approach of Satterthwaite (1946) as developed for cluster-robust meta-regression by Bell and McCaffrey (2002), Tipton (2015), and Pustejovsky and Tipton (2018). For coefficient k, we define the CR2-adjusted projection vectors

$$\mathbf{g}_{jk} \;=\; \mathbf{A}_j^{\mathsf{T}} \mathbf{W}_j \mathbf{X}_j \mathbf{Q}\, \mathbf{c}_k, \qquad j = 1, \ldots, J, \tag{D.1}$$

where $\mathbf{c}_k$ denotes the kth standard basis vector, so that the CR2 variance of $\hat{\beta}_k$ is the quadratic form $[\mathbf{V}_{\mathrm{CR2}}]_{kk} = \sum_j (\mathbf{g}_{jk}^{\mathsf{T}} \hat{\mathbf{e}}_j)^2$. Under the working model, the sampling variability of this quadratic form is governed by the $J \times J$ matrix $\mathbf{P}_k$ with entries

$$[\mathbf{P}_k]_{jl} \;=\; \mathbf{g}_{jk}^{\mathsf{T}}\, \mathrm{Cov}(\hat{\mathbf{e}}_j, \hat{\mathbf{e}}_l)\, \mathbf{g}_{lk}, \tag{D.2}$$

where the residual covariances are evaluated at the working covariance blocks $\boldsymbol{\Sigma}_j$ (plug-in variance components) and account for the hat-matrix structure of Appendix B.4. Matching the first two moments of the quadratic form to a scaled χ^2 yields the Satterthwaite degrees of freedom

$$\mathrm{df}_k \;=\; \frac{\big[\mathrm{tr}(\mathbf{P}_k)\big]^2}{\mathrm{tr}(\mathbf{P}_k^2)}, \tag{D.3}$$

and the CCC-t interval $\hat{\beta}_k \pm t_{1-\alpha/2,\,\mathrm{df}_k}\ \mathrm{SD}(\beta_k^*)$. The construction is deterministic given the data and the plug-in variance components; it is computed once per fit, alongside the target.

D.2. **Per-coefficient variation in the degrees of freedom and its consequences.** No quantity in (D.1)–(D.3) is shared across coefficients, since each $\mathbf{g}_{jk}$ projects a different column of $\mathbf{Q}$. Coefficients whose information is concentrated in a few clusters (moderator slopes identified by between-study contrasts) yield $\mathbf{P}_k$ matrices dominated by a small number of large diagonal entries, and hence small

df_k. In contrast, the intercept spreads its information over all J clusters and has large df_k. In the simulation (Appendix E.6) the median df is 13.4 for the average effect, compared with 4.3–6.0 for the slopes under identical conditions. The practical consequence is developed in the main text (Section 4.3). A normal-reference interval for a coefficient with df $\approx$ 4 is roughly 28% too short even when the standard error is perfectly calibrated. This is the dominant mechanical explanation for the CCC-t–CCC-q coverage gap on slopes, whose ordering across coefficients (0.070, 0.040, 0.037, 0.016) is monotonically related to the df deficit.

D.3. **Plug-in parity with `clubSandwich`.** The `bayesRVE` implementation of (B.10) and (D.3) was validated against the `clubSandwich` package (Pustejovsky, 2026) by supplying identical variance-component plug-ins to both. Agreement is within 10^{-6} for every CR2 standard error and every Satterthwaite df across the validation designs. In routine use the two implementations differ only through their plug-ins: `bayesRVE` evaluates the weights, bread, and $\mathbf{P}_k$ at posterior means, whereas REML-based pipelines evaluate them at REML estimates. Comparisons of df across implementations (e.g., Table 3 of the main text and Appendix F.5) therefore reflect genuine prior-data interaction in the variance components rather than implementation divergence.

Appendix E. Simulation study: complete protocol and full results

E.1. **Aims and overview.** The simulation evaluates the frequentist operating characteristics of Three-Layer Calibration against the frequentist standard of care, an uncalibrated Bayesian baseline, and a normal-reference variant that isolates the tail layer, over a factorial design anchored to a real meta-analytic database. The primary metric is coverage of nominal 95% intervals for the average effect (β_0) and for three moderator slopes (β_1 small, β_2 substantial, β_3 null). The secondary metrics are bias, interval width, RMSE, rejection rates, standard-error calibration, mechanism diagnostics, convergence, and computation time. The estimators are cataloged in Appendix A.2; every estimator analyzes every generated dataset, so all contrasts are within-dataset.

E.2. **Data-generating process.** Each replication generates a meta-analytic dataset in seven steps:

(1) We draw J studies with replacement from the Tanner-Smith–Lipsey (TSL15) study pool (Tanner-Smith & Lipsey, 2015) as curated by Pustejovsky and Tipton

(2022). Each sampled study retains its cluster size (capped at 12 effect sizes), its moderator values, and its sampling-variance profile. In the balanced designs the cluster sizes are overridden by $n_j \equiv 5$ or $n_j \equiv 1$.

(2) We select the moderators. For $K = 1$ the model contains the intercept only. For $K = 4$ it contains the intercept plus three study-level moderators (centered percent college, percent male, attrition). For $K = 7$ it contains the $K = 4$ set plus three within-study outcome-category indicators.

(3) We fix the true coefficients at the values estimated from the TSL15 database (CHE REML fit): for $K = 4$, $\beta_0 = 0.1475$, $\beta_1 = 0.0310$, $\beta_2 = 0.2896$, $\beta_3 = 0$; for $K = 7$, additionally $\beta_4 = 0.0236$, $\beta_5 = 0.0046$, $\beta_6 = 0.0122$ (and $\beta_0 = 0.1405$, $\beta_1 = 0.0295$, $\beta_2 = 0.2898$); for $K = 1$, $\beta_0 = 0.1346$. The empirical REML anchors for the variance components are $\tau = 0.105$ and $\omega = 0.118$, and these values are bracketed by the $\tau \in \{0.1, 0.2\}$ and $\omega \in \{0, 0.1, 0.2, 0.3\}$ grids.

(4) We draw random effects $u_j \sim \mathsf{N}(0, \tau^2)$ between studies and, under CHE generation, $w_{ij} \sim \mathsf{N}(0, \omega^2)$ between effects. In the contaminated-normal sensitivity block the random effects are drawn from a 90/10 mixture in which the variance of the contaminating component is inflated ninefold.

(5) We draw study-specific sampling correlations $\rho_j \sim \text{Beta}(\rho_{\text{true}}\nu, (1 - \rho_{\text{true}})\nu)$ with $\nu = 9$ (degenerate at 0 when $\rho_{\text{true}} = 0$), so that the true dependence varies across studies around the nominal level; the realized ρ_j is not supplied to any estimator.

(6) We generate correlated standardized mean differences by simulating each study's raw mean-difference vector from $\mathsf{N}(\boldsymbol{\delta}_j, (4/N_j)\boldsymbol{\Sigma}_j^{\rho_j})$ with equicorrelation $\boldsymbol{\Sigma}_j^{\rho_j} = (1-\rho_j)\mathbf{I}+\rho_j\mathbf{J}$, by taking the pooled covariance from a Wishart$(N_j - 2, \boldsymbol{\Sigma}_j^{\rho_j})$ draw, and by applying Hedges' small-sample correction $J_{\text{corr}} = 1 - 3/\{4(N_j - 2) - 1\}$ to obtain g and its sampling variance $v = J_{\text{corr}}^2(4/N_j) + g^2/\{2(N_j - 2)\}$.

(7) We rejection-sample until the realized design matrix has full column rank and, for $K = 7$, each outcome-category indicator appears in at least two studies; Wishart degrees-of-freedom guards cap the cluster sizes where needed.

The generating process therefore embeds the three features that motivate calibration: sampling errors that are genuinely correlated, with correlations that are unknown and vary across studies; cluster sizes that are unbalanced and empirically realistic; and moderators whose information is concentrated in few clusters.

E.3. **Condition grid.** The crossed factors are the generating structure (CE, CHE); the number of studies $J \in \{5, 10, 15, 20, 40, 80\}$; the number of coefficients $K \in \{1, 4, 7\}$, with $K = 7$ restricted to $J \geq 16$ at the design stage; $\tau \in \{0.1, 0.2\}$; $\omega \in$

$\{0, 0.1, 0.2\}$ (0 under CE only; 0.3 in a sensitivity block); $\rho_{\text{true}} \in \{0, 0.3, 0.8\}$ (0.6 in a sensitivity block); and the cluster-size profile (TSL15-empirical, balanced-5, balanced-1). After removal of structurally infeasible and redundant cells, the design retains 324 conditions. The 282 core and balanced conditions are run with 1,000 replications each, and the 42 sensitivity-block conditions (contaminated random effects; extreme $\omega = 0.3$; $\rho_{\text{true}} = 0.6$; $\rho_{\text{work}} = 0.6$; narrow and diffuse priors; balanced-1) are run with 500 replications each. The frequentist comparators assume working correlation $\rho_{\text{work}} = 0.8$ in the core grid and 0.6 in the corresponding sensitivity block. The full machine-readable grid is distributed with the simulation codebase (Appendix G).

E.4. **Estimation settings and convergence gates.** Bayesian fits use the `bayesRVE` defaults: weakly informative normal priors on $\boldsymbol{\beta}$, half-$t_3(0, 0.5)$ priors on τ and ω (scale 0.1/2.0 in the narrow/diffuse sensitivity block), a non-centered parameterization for CE, and Stan sampling with 4 chains $\times$ (1,000 warmup + 1,000 sampling) = 4,000 posterior draws at `adapt_delta` = 0.95. A replication is classified as *converged* if $\hat{R}_{\max} \leq 1.05$, bulk ESS ≥ 400 (with values in $[100, 400)$ classified as marginal), tail ESS ≥ 200, divergence rate $\leq 1\%$, and treedepth-hit rate $\leq 5\%$; failures are retried once at `adapt_delta` = 0.99 with 2,000 warmup iterations. Converged and marginal fits enter the analysis.

The headline results use the *converged-only* policy within the *operating envelope* $J \geq J_{\min} = \max(8, 2K + 2)$. The CE-arm CCC-$t$ refuses to run below the envelope ($J = 5$ with $K \geq 4$); this refusal is a designed applicability boundary, and within the envelope its median convergence rate is 1.000. The analyzed bases are consequently as follows: for β_0, 324 design conditions, of which 319 are in-envelope; for the slopes β_1–β_3, 177 correctly specified $K \geq 4$ conditions (59 CE, 118 CHE), of which 176 are in-envelope; and for the higher-order coefficients β_4–β_6, 54 conditions ($K = 7$; 18 CE, 36 CHE). A worst-case sensitivity analysis recodes every refused or non-converged replication as a miss; it does not change any qualitative conclusion, and the complete condition-level convergence table is provided with the result files.

E.5. **Full coverage results.** Table E.1 expands Table 2 of the main text by working-model arm. Figure E.1 displays coverage for β_0 by method, Figure E.2 maps slope coverage over the design, and Figure E.3 traces small-slope coverage as a function of J, which isolates the Layer-3 effect at every design size.

Table E.1. Coverage of nominal 95% intervals by method, arm, and estimand (correctly specified, in-envelope, converged-only; unweighted means over conditions, with between-condition standard errors in parentheses for the slope headline rows). The Pooled column is the unweighted mean over the union of CE and CHE conditions and matches Table 2 of the main text exactly. The external `robumeta` benchmark (M9; CE, $K = 1$ only, 51 conditions) attains 0.946 for β_0.

Estimand	Method	CE arm	CHE arm	Pooled
β_0	CCC-t	0.952	0.952	0.952
	CR2	0.946	0.948	0.947
	CCC-q	0.937	0.935	0.936
	Uncalibrated Bayes	0.953	0.952	0.952
β_1	CCC-t	0.978 (0.002)	0.978 (0.001)	0.978
	CR2	0.906 (0.007)	0.904 (0.005)	0.905
	CCC-q	0.909 (0.003)	0.907 (0.002)	0.908
	Uncalibrated Bayes	0.961	0.961	0.961
β_2	CCC-t	0.971 (0.002)	0.970 (0.001)	0.971
	CR2	0.961 (0.001)	0.961 (0.001)	0.961
	CCC-q	0.932 (0.002)	0.931 (0.001)	0.931
	Uncalibrated Bayes	0.961	0.960	0.960
β_3	CCC-t	0.967 (0.002)	0.966 (0.001)	0.966
	CR2	0.954 (0.002)	0.956 (0.001)	0.955
	CCC-q	0.930 (0.003)	0.930 (0.001)	0.930
	Uncalibrated Bayes	0.970	0.962	0.965

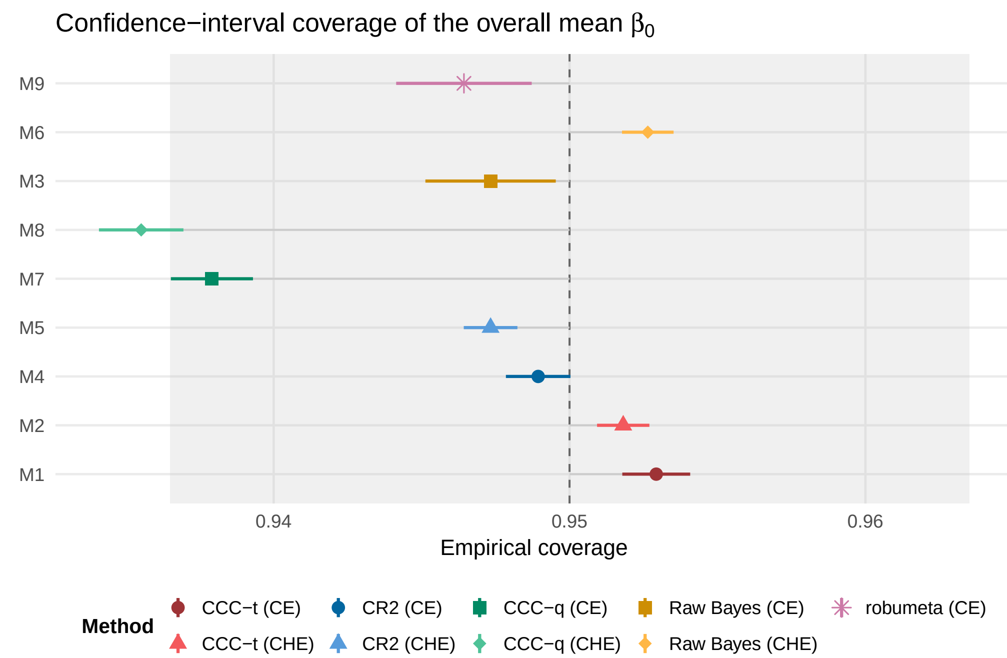

FIGURE E.1. Coverage of nominal 95% intervals for the average effect β_0 by method (in-envelope base, 319 conditions). All method families perform adequately for this estimand; the differences between methods appear on the slopes.

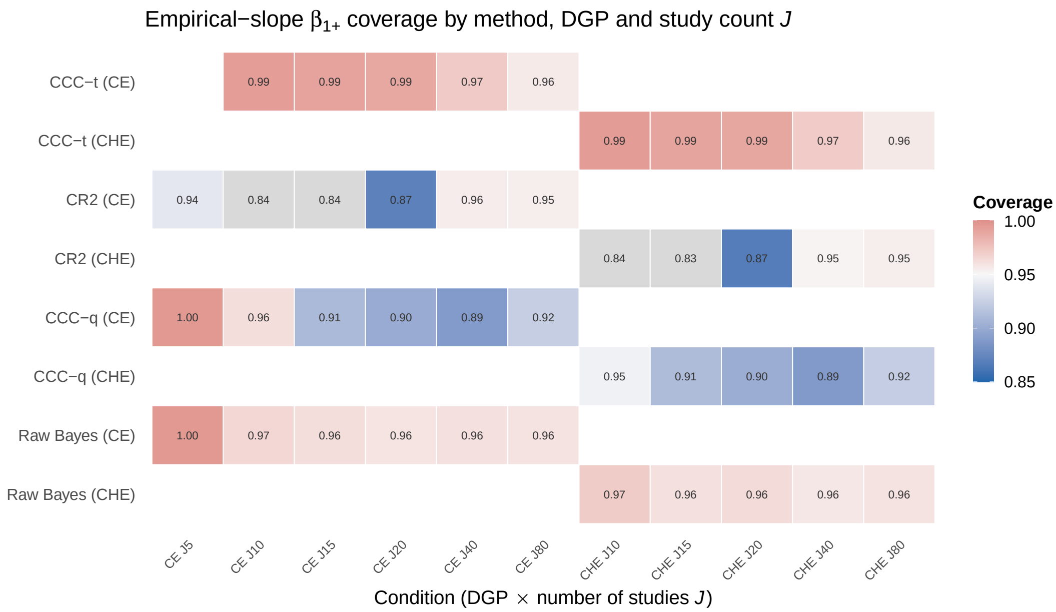

FIGURE E.2. Slope coverage across the design grid (heatmap over J and generating structure, by method and coefficient). The coverage deficits of CR2 and CCC-q on β_1 are not confined to a small region of the design.

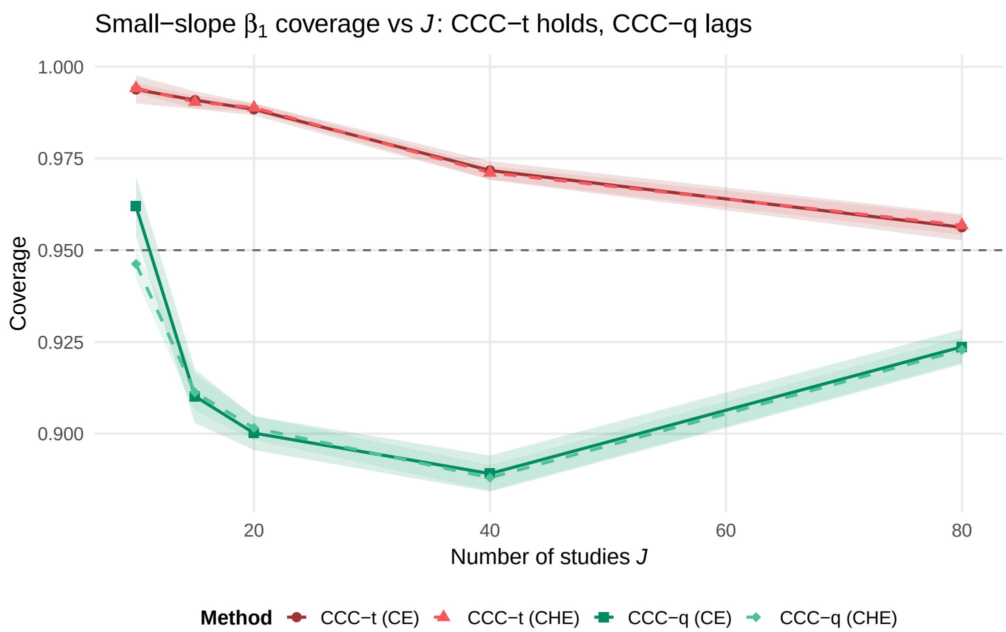

FIGURE E.3. Small-slope (β_1) coverage as a function of J for CCC-t and CCC-q on identical calibrated draws. The gap narrows only slowly with J because the Satterthwaite df for the slope grow with the number of informative clusters rather than with J alone.

E.6. **The Layer-3 mechanism: per-coefficient degrees of freedom.** Table E.2 gives the detail behind Figure 3 of the main text, with coverage and median Satterthwaite df presented side by side for each arm. Pooled across the finite-df methods on the common $K \geq 4$ base, the median dfs are 4.25 (β_1), 4.93 (β_2), 5.99 (β_3), and 13.44 (β_0), so the slopes carry 32–45% of the intercept's df. The paired per-condition CCC-t−CCC-q coverage gaps are 0.070, 0.040, 0.037, and 0.016 respectively; the gaps are therefore monotone in df across all four coefficients.

TABLE E.2. Layer-3 comparison of coverage and median Satterthwaite df by arm (CCC-t rows), with CCC-q computed on identical calibrated draws but evaluated against a normal reference (df $= \infty$).

Coefficient	Method (arm)	Conditions	Coverage	Median df
β_1	CCC-t (CE)	58	0.978	4.37
	CCC-t (CHE)	118	0.978	4.31
	CCC-q (CE)	58	0.909	∞
	CCC-q (CHE)	118	0.907	∞
β_2	CCC-t (CE)	58	0.971	6.46
	CCC-t (CHE)	118	0.970	6.39
	CCC-q (CE)	58	0.932	∞
	CCC-q (CHE)	118	0.931	∞
β_3	CCC-t (CE)	58	0.967	8.03
	CCC-t (CHE)	118	0.966	8.26
	CCC-q (CE)	58	0.930	∞
	CCC-q (CHE)	118	0.930	∞

E.7. **Where the uncalibrated interval fails.** Averaged over correctly specified conditions, the uncalibrated credible interval appears adequate (Table E.1). The failure is confined to the tail of the distribution over conditions and arises under working-model misspecification, that is, when the CE working model is fitted to CHE-generated data. Under that specification role, uncalibrated-Bayes coverage for β_0 averages 0.944 but reaches a minimum of 0.818, with the worst cells at $\omega = 0.2$ and large J. Coverage decreases as studies accumulate (0.843 at $J = 20$, 0.826 at $J = 40$, 0.818 at $J = 80$; all at $K = 7$, $\tau = 0.1$, empirical cluster sizes), because additional data sharpen the posterior around a working-model spread that is incorrect. On the identical cells, CCC-t averages 0.952–0.954 with a minimum of 0.932 across the entire misspecified block. (These values are derived from the archived condition-level results by the same pooling procedure as all other cells; see Appendix G.) These results give operational content to the claim in the main text that the uncalibrated interval is adequate exactly when the working model is adequate, a condition that the analyst cannot verify from within the model.

E.8. **Type-I error and power.** Table E.3 collects the rejection rates. CCC-t controls the Type-I error for the null slope at 0.033–0.034, compared with 0.044–0.046 for CR2. CCC-q inflates the Type-I error to 0.070, which is the testing counterpart of its undercoverage, and CCC-q similarly over-rejects on the substantial slope (0.284), so its apparent power is not usable. The cost of the proposed method appears in the β_1 row, where the power is 0.027 for CCC-t against 0.101 for CR2.

TABLE E.3. Rejection rates (slope base 177; β_0 base 324). The power rows have nonzero true values, and the β_3 row gives the Type-I error at nominal 0.05. Parenthetical entries are (CE arm / CHE arm).

	CCC-t	CR2	CCC-q
Type-I, $\beta_3 = 0$	0.034 (0.033 / 0.034)	0.045 (0.046 / 0.044)	0.070
Power, $\beta_1 = 0.031$	0.027	0.101	0.100
Power, $\beta_2 = 0.290$	0.195 (0.215 / 0.185)	0.208	0.284
Power, $\beta_0 = 0.148$	0.788 (CE)	—	—

E.9. **Efficiency, width, and bias.** Table E.4 reports the interval-scale efficiency metrics, and Figures E.4 and E.5 display the two headline patterns. Median widths on β_2 favor CCC-t (1.11 against 1.35 for CR2 in the CE arm), and relative RMSE also favors it (0.79 CE, 0.90 CHE); both patterns reflect the shrinkage of the Bayesian point estimate. On β_1 the ordering is reversed by a large margin: the CCC-t median width is approximately 958 (CE) and 1,118 (CHE) against approximately 0.95–0.99 for CR2, a difference of three orders of magnitude. The reason is that the small slope carries almost no likelihood information, so the posterior for it is wide and heavy-tailed, and the near-floor Satterthwaite df multiply the quantile. This width produces the coverage of 0.978 and the power of 0.027, and both quantities are reported in the main text. Bias is small throughout: CCC-t is near-unbiased on β_1 (-0.004 CE, -0.003 CHE) and β_3, with modest shrinkage on β_2 (-0.031 CE, -0.029 CHE; about 10% of the true 0.290), against CR2 values of -0.001 to -0.004. The bias-eliminated coverage in the archived tables shows that the slope-coverage conclusions are not artifacts of bias.

TABLE E.4. Efficiency metrics on the slope base (medians over conditions). Relative RMSE is RMSE(CCC-t)/RMSE(CR2) within arm; ratio metrics are reported only for the scale-safe β_2 (Appendix E.10).

	Median CI width		Relative RMSE vs. CR2	
Coefficient	CCC-t (CE / CHE)	CR2 (CE / CHE)	CE	CHE
β_1	958 / 1,118	0.95 / 0.99	0.89	—
β_2	1.11 / —	1.35 / —	0.79	0.90
β_3	0.77 / —	0.86 / —	0.83	—

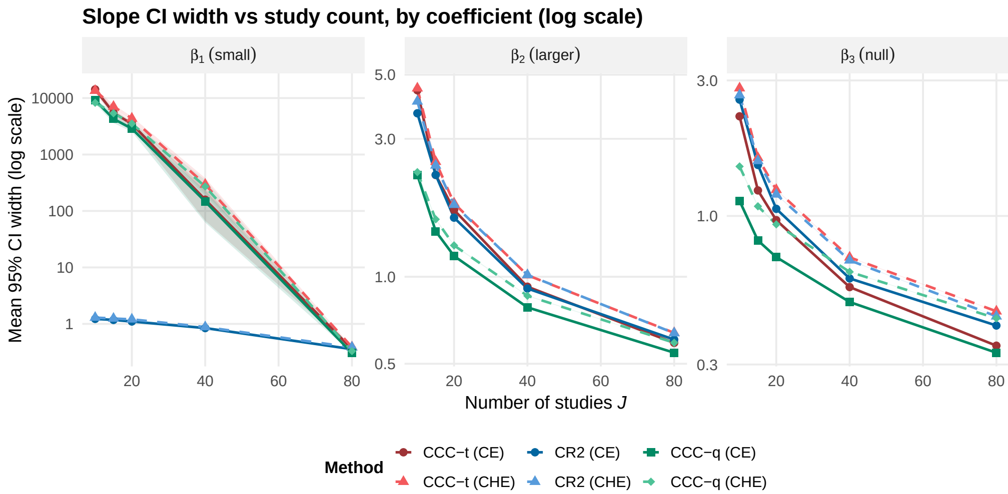

FIGURE E.4. Median interval width for β_1 by J (log scale), showing two distinct width regimes. The CCC-t width is driven by the weakly identified posterior and the small-df tails, whereas the CR2 width is driven by the sandwich SE.

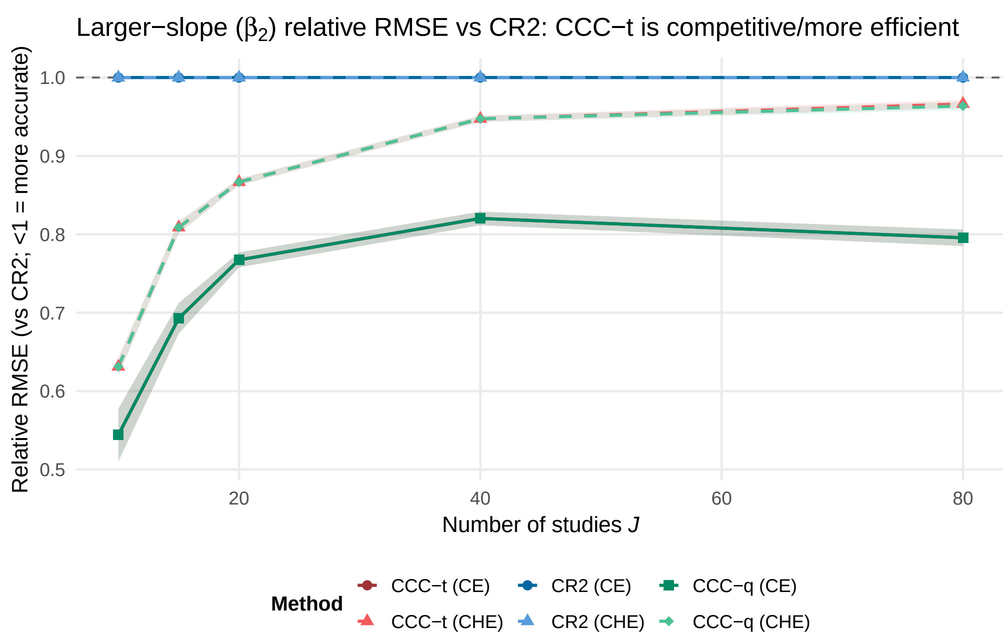

FIGURE E.5. Relative RMSE (CCC-t/CR2) for the substantial slope β_2 by J. The efficiency gain from the shrinkage of the Bayesian point estimate is largest at small J.

E.10. **Standard-error calibration and the scale-degeneracy policy.** On the scale-safe β_2, the median relative SE error (model SE against empirical SE) is +6.7% (CE) and +8.3% (CHE) for CCC-t, +6.9% for CCC-q (CE), and −4.9% (CE) and −1.6% (CHE) for CR2; for β_0 the CCC-t values are +0.7% and +0.6%. The proposed intervals are therefore mildly conservative in SE terms on the slopes and nearly exact on the average effect, whereas CR2 is mildly anticonservative.

Ratio-based metrics are not informative when the true coefficient is near zero. On β_1 (true value 0.031) the relative SE error metric reaches approximately $1.7 \times 10^6\%$ and the calibration-ratio summary reaches ≈ 595 solely because the denominators are near zero; the values are numerically well defined but carry no inferential content (Figure E.6). The pre-declared reporting policy is therefore to report ratio metrics (R_k, relative SE error, relative precision) for β_0 and β_2 only, and interval-scale metrics (coverage, width, rejection) for all coefficients. On the scale-safe set, the R_k medians are 0.94–0.96 (β_0) and 0.97 (β_2), so CCC typically shrinks the posterior spread slightly toward the target.

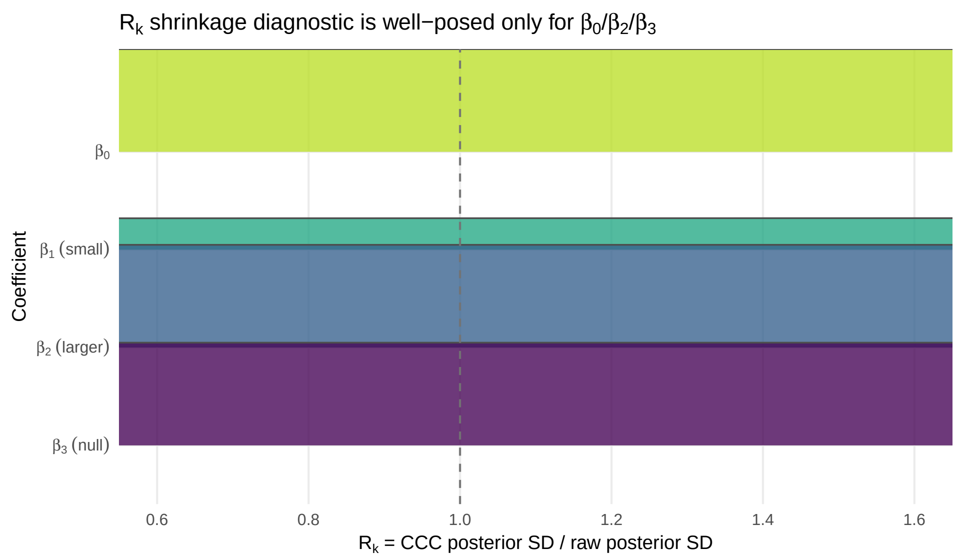

FIGURE E.6. Scale degeneracy of ratio metrics for coefficients with near-zero true values. The same estimator that is well calibrated on interval-scale metrics produces meaningless ratio summaries on β_1, which motivates the scale-safe reporting policy.

E.11. **Mechanism diagnostics.** On the diagnostic subset with archived draw-level output (93 conditions), the calibration-matrix deviation $\|\mathbf{A}_{\text{ccc}} - \mathbf{I}_K\|_F$ has a median of 0.32 with a heavy right tail (the means are in the hundreds); the tail comprises the conditions in which the calibration makes its largest corrections. The deviation is identical across coefficients by construction, since there is one matrix per fit. The CCC-t $\boldsymbol{\beta}$-block skip rate is 0.000 in both arms; the CCC-q CE variant occasionally skips (mean rate 0.054). The Theorem-2 operating ratio, defined as the calibrated SD divided by the CR2 SE on β_2, is 0.897 (CE) and 1.017 (CHE). Under CE the calibrated Bayesian spread therefore lies below the frequentist benchmark while covering better. This combination is an efficiency gain inherited from the posterior geometry rather than a paradox, since the coverage is carried by the df-aware tails.

E.12. **Sensitivity blocks.** Figures E.7 to E.11 summarize the sensitivity analyses. Under working-model misspecification the slope conclusions are unchanged: CCC-t attains 0.977–0.978 on β_1 across the correct, overfit, and misspecified roles, against 0.905–0.919 for CR2. Contaminated (heavy-tailed) random effects and extreme within-study heterogeneity ($\omega = 0.3$) leave the ordering intact on their small feasible slope bases (2 and 4 conditions per method, respectively); these blocks are reported as directional evidence rather than as precise estimates. The prior-sensitivity and working-correlation blocks are intercept-only by design ($K = 1$), so no slope evidence exists there, and extending them is left to future work. Across priors (half-t_3 scale 0.1/0.5/2.0) and $\rho_{\text{work}} \in \{0.6, 0.8\}$, the β_0 conclusions are stable.

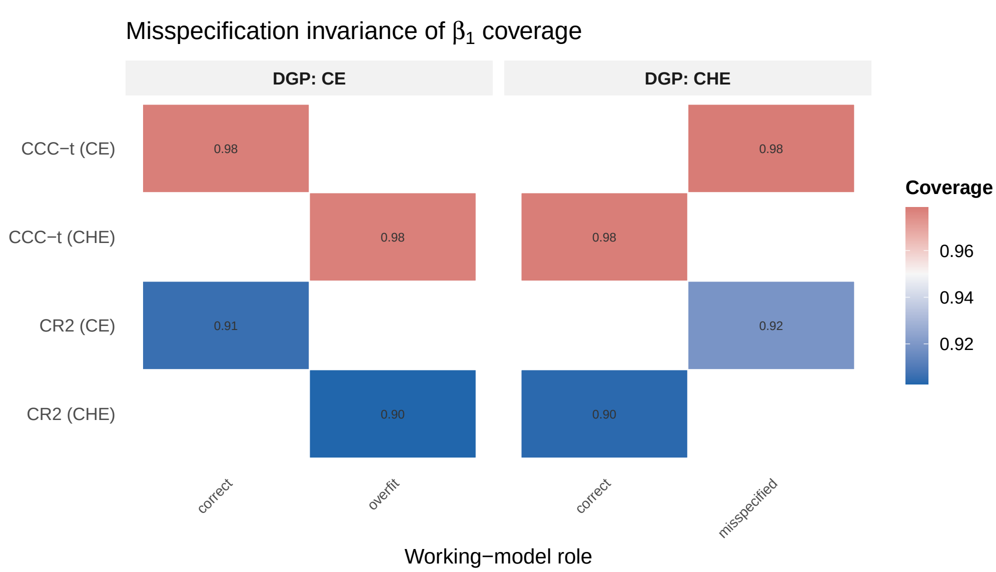

FIGURE E.7. Small-slope coverage under working-model misspecification (CE methods on CHE data) across the design. The CCC-t advantage is preserved.

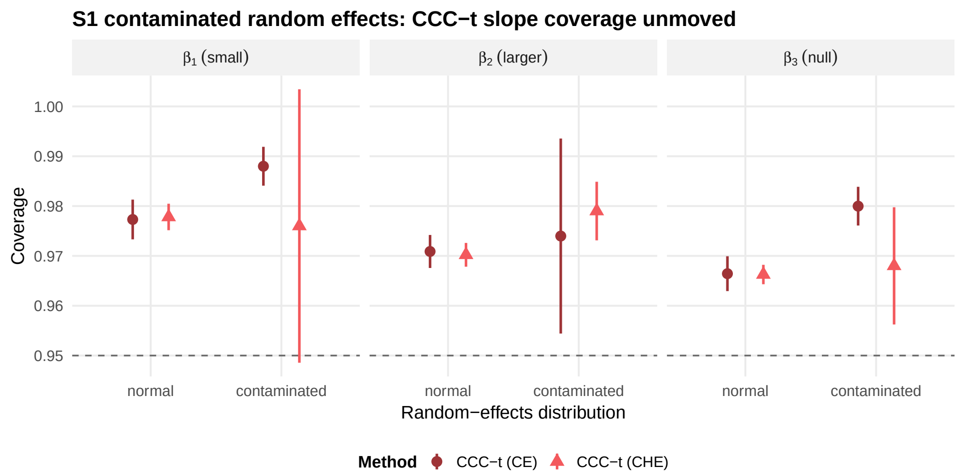

FIGURE E.8. Contaminated random effects (10% contamination, variance $\times 9$; sensitivity block). The slope coverage ordering is unchanged; the condition base is small, so the evidence is directional.

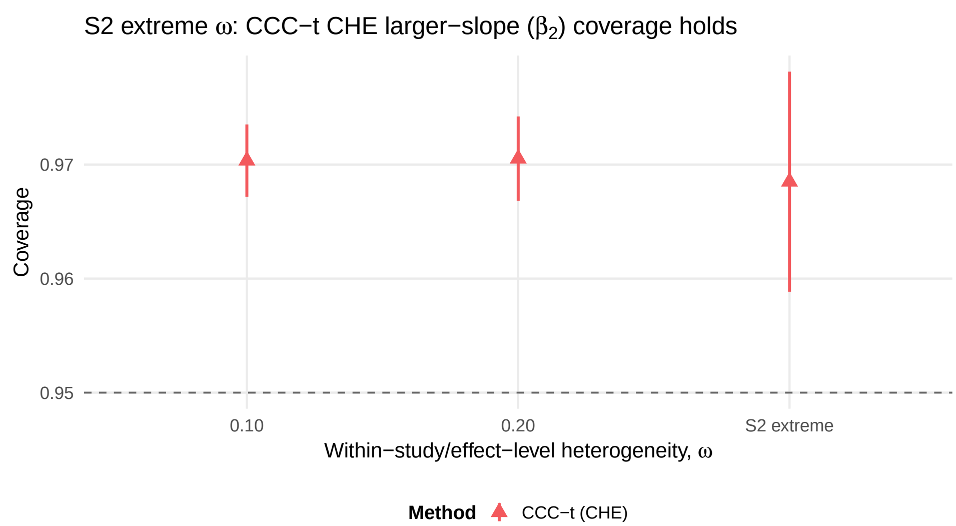

FIGURE E.9. Extreme within-study heterogeneity ($\omega = 0.3$; sensitivity block). The slope coverage ordering is unchanged; the condition base is small, so the evidence is directional.

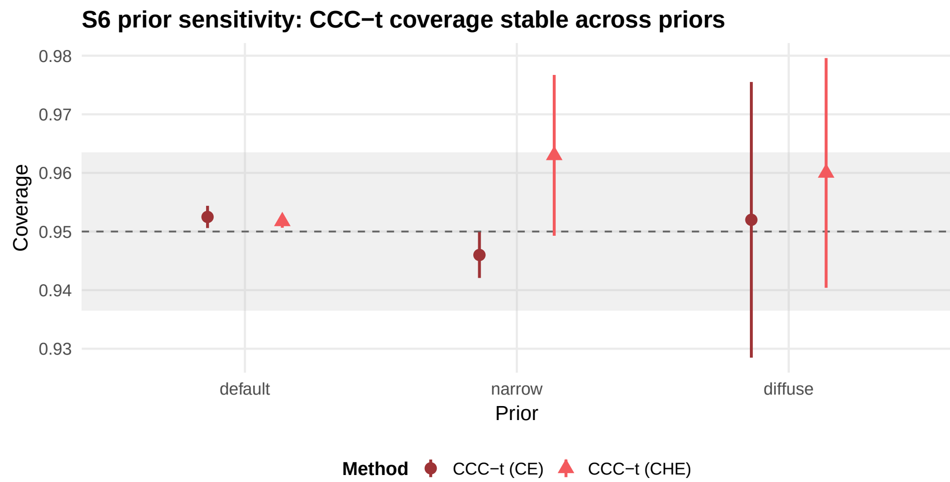

FIGURE E.10. Prior sensitivity (β_0; half-t_3 scales 0.1, 0.5, 2.0). Calibrated coverage is stable across prior choices.

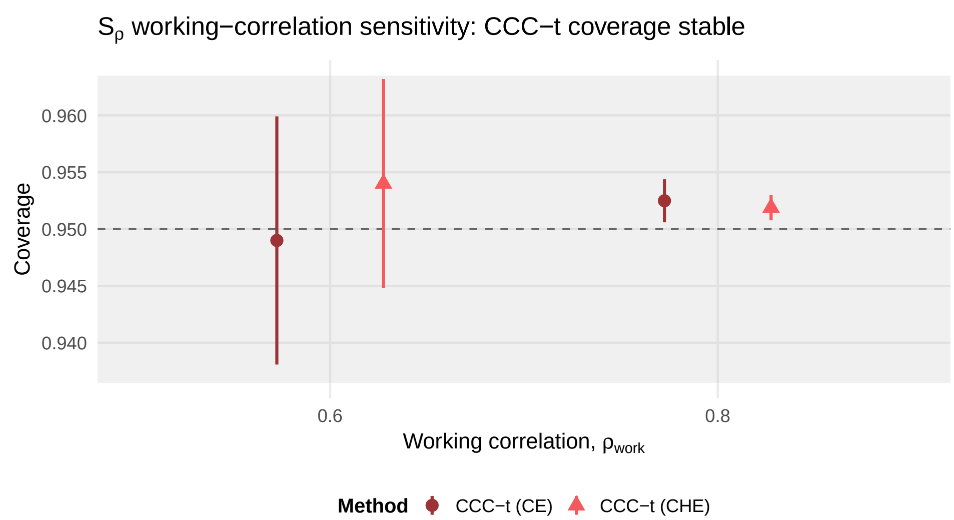

FIGURE E.11. Working-correlation sensitivity (β_0; $\rho_{\text{work}} = 0.6$ versus 0.8 in the frequentist comparators). The conclusions are stable.

E.13. **Convergence and computation.** Figure E.12 maps convergence over the grid. Within the envelope, the Bayesian arms converge in effectively all replications (median rate 1.000); the below-envelope CE cells at $J = 5$, $K \geq 4$ are refused by design. The median computation time per replication is 19.3 s for CCC-t in the CE arm and 9.6 s for CCC-t in the CHE arm, essentially all of which is MCMC sampling, against 0.26 s for CR2; the calibration layers themselves take less than a millisecond.

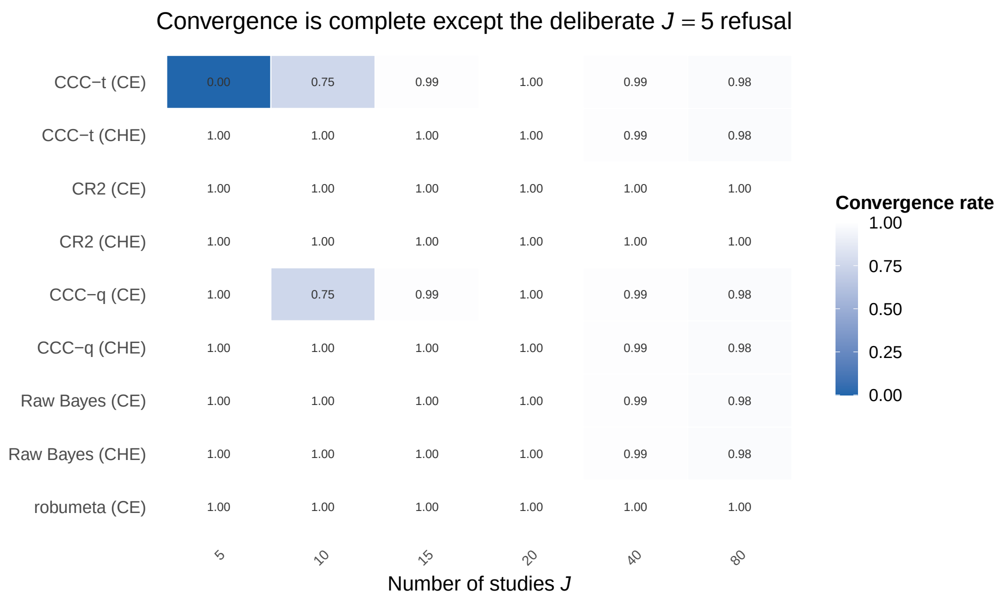

FIGURE E.12. Convergence rates over the condition grid. The only structural non-convergence is the designed refusal below the operating envelope $J_{\min} = \max(8, 2K + 2)$.

Appendix F. Empirical illustration: complete TSL15 results

F.1. **Data and specification.** The empirical analysis uses the Tanner-Smith and Lipsey meta-analysis of brief alcohol interventions for adolescents and young adults (Tanner-Smith & Lipsey, 2015), in the analytic form curated by Pustejovsky and Tipton (2022). The dataset comprises $J = 117$ studies and $N = 1{,}198$ standardized mean differences (Hedges' g), with 1–108 effect sizes per study (median 6). The effect sizes fall into six outcome categories (frequency of use, frequency of heavy use, quantity of use, peak consumption, blood alcohol concentration, combined measures), and five continuous moderators are available (centered follow-up week, long-follow-up indicator, percent college, percent male, attrition).

Both working models were fitted to the cell-means meta-regression (outcome-category means plus the five moderators, $K = 11$). The priors were $\beta_k \sim \mathsf{N}(0, 10^2)$ and $\tau, \omega \sim$ half-Cauchy$(0, 0.5)$, the choice adopted for the empirical analysis of this dataset; the half-Cauchy is the $\nu = 1$ member of the half-t family that the package uses by default (Appendix G.1). Each fit used 4 chains $\times$ (2,000 warmup + 2,000 sampling) = 8,000 draws, a fixed seed of 20260324, `adapt_delta` = 0.95, and `max_treedepth` = 12. The convergence diagnostics were $\hat{R}_{\max} = 1.008$ (CE) and 1.005 (CHE), a minimum bulk ESS of 415, a minimum tail ESS of 1,109, and zero divergent transitions. The frequentist benchmark fits the same structures in `metafor`, with sampling covariances imputed under equicorrelation $\rho = 0.6$ and CR2 standard errors with Satterthwaite df from `clubSandwich`; `robumeta` provides the classic correlated-effects RVE fit for reference.

F.2. **CHE working model: outcome-category means.** Table 3 of the main text reports the CE comparison; Table F.1 reports the corresponding CHE comparison.

TABLE F.1. TSL15, CHE working model: outcome-category mean effects (adjusted for the five moderators) under the frequentist standard of care and Three-Layer Calibration.

	CR2 (frequentist)			Three-Layer Calibration			
Outcome category	Est.	SE	df	Est.	Calib. SD	df	r_k
Frequency of use	0.118	0.027	53.2	0.131	0.027	67.4	1.008
Frequency of heavy use	0.170	0.026	57.6	0.173	0.028	70.9	1.049
Quantity of use	0.151	0.023	75.6	0.158	0.024	91.7	0.977
Peak consumption	0.154	0.033	27.1	0.155	0.031	31.1	1.078
Blood alcohol concentration	0.181	0.029	38.9	0.183	0.028	42.6	0.934
Combined measures	0.160	0.036	24.3	0.163	0.033	27.4	0.978

F.3. **Calibration ratios and the acceptance gate.** The validation gate prespecified for this analysis requires the ratio of calibrated SD to CR2 SE to lie in $[0.9, 1.1]$ for every outcome-category coefficient. Table F.2 lists all twelve ratios, and every coefficient passes under both working models. The CE ratios (mean 0.9976) are more tightly clustered than the CHE ratios (mean 0.9953, range 0.9250–1.0544); the greater dispersion of the CHE ratios reflects the additional variance component in the plug-ins. Both means lie within half a percent of unity.

Table F.2. Per-coefficient calibration ratios (calibrated SD / CR2 SE) for the CE and CHE working models.

Outcome category	CE	CHE
Frequency of use	1.028	1.033
Frequency of heavy use	0.992	1.054
Quantity of use	1.014	1.046
Peak consumption	0.981	0.941
Blood alcohol concentration	0.979	0.972
Combined measures	0.991	0.925
Mean	0.998	0.995
Range	[0.979, 1.028]	[0.925, 1.054]
Gate $[0.9, 1.1]$	pass (6/6)	pass (6/6)

F.4. **Variance components.** Table F.3 compares variance-component estimates across estimation frameworks. The Bayesian posterior means of τ exceed the REML estimates, a pattern typical of a heavy-tailed half-Cauchy prior on a scale parameter, whereas the posterior mean of ω is below the REML estimate; the two components trade off within the nearly unconfounded joint posterior (posterior correlation ≈ -0.04 at $J = 117$). These differences are expected and documented consequences of the priors rather than calibration artifacts. Layer 2 calibrates the spread of $\boldsymbol{\beta}$ to the CR2 target constructed at the Bayesian plug-ins, and the empirical gate of Appendix F.3 shows that the result nevertheless matches the REML-based frequentist target to within a few percent.

Table F.3. Variance-component estimates for TSL15 (outcome-category model).

Model (framework)	τ	ω
CE, `robumeta`	0.182	—
CE, `metafor` REML	0.141	—
CHE, `metafor` REML	0.123	0.116
CE, `bayesRVE` posterior mean	0.222	—
CHE, `bayesRVE` posterior mean	0.219	0.061

F.5. **Satterthwaite degrees of freedom across implementations.** The per-coefficient dfs in Table 3 of the main text and Table F.1 differ between `bayesRVE` and `metafor`/`clubSandwich` (e.g., 35.9 versus 51.4 for peak consumption under CE). The difference is entirely a plug-in effect. The `bayesRVE` implementation evaluates the weight matrices, bread, and df quadratic forms at posterior-mean variance components, whereas the frequentist pipeline evaluates them at REML estimates, and the discrepancy in $\hat{\tau}$ (Table F.3) propagates through the leverage structure. When identical plug-ins are supplied, the two implementations agree to 10^{-6} (Appendix D.3). At the df magnitudes of this dataset (> 24 for every coefficient), the practical effect on interval endpoints is negligible, since the t multipliers differ only in the third decimal place.

F.6. **Control-condition model.** As a robustness check, a parallel specification replacing the six outcome-category means with four control-condition means ($K = 9$) was fitted under both working models. Its MCMC diagnostics are comparable to those of the outcome-category models, and its variance components are in line with Table F.3 (CE $\hat{\tau} = 0.217$; CHE $\hat{\tau} = 0.215$, $\hat{\omega} = 0.060$). Full numerical output for both specifications is included in the replication materials (Appendix G).

Appendix G. Software and reproducibility

G.1. **The `bayesRVE` package.** Three-Layer Calibration is implemented in the open-source R package `bayesRVE` (Lee & Tipton, 2026) (v0.7.0 at the time of writing; MIT license), with Stan (Carpenter et al., 2017) as the sampling back end. The development repository is available at https://github.com/joonho112/bayesRVE, and the documentation site at https://joonho112.github.io/bayesRVE/. A complete analysis can be carried out as follows.

Listing G.1. Minimal `bayesRVE` workflow.

```r
library(bayesRVE)

fit <- bm_fit(yi = yi, vi = vi, X = X, study = study,
              model = "che")                # or "ce"
sw  <- sandwich(fit, rho = 0.6, small_sample = "CR2")
cal <- ccc(fit, sw)                          # blockwise CCC
summary(cal)                                 # CCC-t intervals, df_k, r_k
```

The function `bm_fit()` fits the CE model (non-centered parameterization by default; a centered variant is available) or the CHE model (marginal likelihood, with study effects integrated out), with weakly informative normal priors on $\boldsymbol{\beta}$ and half-t priors on τ and ω as the package default (the half-Cauchy prior used in the empirical analysis of Appendix F is the $\nu = 1$ special case). `sandwich()` builds the CR2 target and the per-coefficient Satterthwaite df at posterior-mean plug-ins, and stores the user's working correlation for the frequentist-comparison and sensitivity utilities (`compare_with_cr2()` and `bm_sensitivity_rho()`). Note that the marginal $\boldsymbol{\beta}$-block target itself is invariant to that stored value. `ccc()` applies the blockwise transformation and records the diagnostics (r_k, $\|\mathbf{A}_{\text{ccc}} - \mathbf{I}\|_F$, and the skip flags). `summary()` and `confint()` report CCC-t intervals by default, with CCC-q and uncalibrated summaries available for comparison. The package enforces the operating envelope $J \geq \max(8, 2K + 2)$ with a warning band near the boundary, and runs the verification suite of Appendix C.4 in its automated tests.

G.2. **Stan models and MCMC configuration.** The package includes three Stan programs: the CE conditional model in non-centered and centered parameterizations, and the CHE marginal model. The generated quantities include study-level log-likelihoods, score contributions, and posterior-predictive draws, which the sandwich construction uses without refitting. The default settings are 4 chains, 2,000 warmup and 2,000 sampling iterations per chain, `adapt_delta` $= 0.95$, and `max_treedepth` $= 12$. The simulation used 1,000+1,000 iterations per chain, with one automatic retry at `adapt_delta` $= 0.99$ and doubled warmup for flagged fits, under the convergence gates of Appendix E.4.

G.3. **Reproducibility of the results in this article.** We describe in turn the simulation study, the empirical analysis, and the archiving of the materials.

The simulation codebase (condition grid, data-generating functions, estimator wrappers M1–M9, metric engine, and post-processing pipeline) is version-controlled alongside the package. The full condition grid is distributed as a machine-readable table, and the complete study documentation (design, verification reports, and full results) is available at https://joonho112.github.io/bayesRVE-simulation-study/. Every figure and table in the main text and in this supplement is generated by scripted builders from a single long-format results file, and each headline number carries a machine-readable provenance record (source file and cell). The analysis layer is bit-reproducible from the archived replication-level results, in the sense that rebuilding the metrics file from the frozen replication data reproduces it exactly (verified by checksum). Exact re-execution of the Monte Carlo sampling itself is reproducible up to the usual floating-point caveats of compiled Stan across tool-chain versions. Session information (R and package versions) is archived with the results.

The TSL15 analysis is fully scripted, with a fixed seed (20260324), an archived data object, and a rendered report containing every table of Appendix F. The dataset derives from the published meta-analysis of Tanner-Smith and Lipsey (2015), as distributed in the replication materials of Pustejovsky and Tipton (2022).

A replication package containing the simulation code, the condition grid, and all result files that reproduce the numbers, tables, and figures of the article is available at https://github.com/joonho112/bayesRVE-replication, with a rendered companion guide at https://joonho112.github.io/bayesRVE-replication/. The merged replication-level results (the analytic bulk) are archived on Zenodo at https://doi.org/10.5281/zenodo.21259722; upon journal acceptance, the replication package itself will additionally be archived with a DOI. The software and data availability statements in the main text record the access points.